DESIGN OF 10 MLD SEWAGE TREATMENT PLANT BASED ON UPFLOW ANAEROBIC SLUDGE BLANKET (UASB) REACTOR WITH DESIGN OF UASB IN MS EXCEL

PROJECT REPORT

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT

FOR THE AWARD OF THE DEGREE OF

Bachelor of Technology

IN

Civil Engineering

By

ARYAN VARSHNEY
SHISHIR

Under the Guidance of

PROF. FARRUKH BASHEER

DEPARTMENT OF CIVIL ENGINEERING

ZAKIR HUSSAIN COLLEGE OF ENGINEERING & TECHNOLOGY

ALIGARH MUSLIM UNIVERSITY

2022-2023

DEPARTMENT OF CIVIL ENGINEERING
ZAKIR HUSSAIN COLLEGE OF ENGINEERING AND TECHNOLOGY
ALIGARH MUSLIM UNIVERSITY
ALIGARH 202001 UP INDIA

CERTIFICATE

This is to certify that the Project entitled "**Design of 10 MLD Sewage Treatment Plant Based on Upflow Anaerobic Sludge Blanket (UASB) Reactor With Design of UASB in MS Excel**" submitted by Aryan Varshney (19CEB518) and Shishir (19CEB116) in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Civil Engineering at **Zakir Husain College of Engineering & Technology, Aligarh Muslim University, Aligarh** is an authentic work carried out by them under my supervision and guidance.

It is further certified that to the best of my knowledge, the matter embodied in the project has not been submitted to any other University/Institute for the award of any Degree or Diploma.

(Prof. Farrukh Basheer)

Asst. Professor

(Department of Civil Engineering)

Aligarh Muslim University

ACKNOWLEDGEMENT

It offers us great pleasure to acknowledge and offer gratitude to all those individuals who have directly or indirectly helped us through the course of this project. This project would never have been completed without the contribution of those people. Unfortunately, the list of acknowledgements, no matter how extensive is always incomplete and inadequate. Indeed, this page of mention shall never be able to touch the horizon of generousness of those who tendered their help to us. We would like to thank AMU Aligarh for giving us the opportunity to use their resources and work in such a challenging environment.

First and foremost, we take this opportunity to express our deepest sense of gratitude to Asst. Prof. Farrukh Basheer for their guidance and immense patience during our project. This project would not have been possible without his help and the valuable time that he has given us amidst his busy schedule.

We would also like to extend our gratitude to all our friends and senior students, who have always encouraged and supported us in doing our work. We would like to thank all the staff Members of Department of Civil Engineering who have always been cooperative with us.

Last but not the least we would like to thank the authors of various research articles and books that we referred to during the course of this project.

ARYAN VARSHNEY **SHISHIR**

DEDICATION

We dedicate this project work to

our beloved parents and specially our

respected teachers who supported

us in the successful completion

of this project

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

CHAPTER

1. INTRODUCTION

1.1 GENERAL

Wastewater is a mixture of water and water-borne wastes generated by residences, commercial and industrial facilities, and organisations. Untreated wastewater often includes high quantities of organic material, pathogenic microbes, nutrients, and toxic substances, resulting in pollution and health risks. As a result, waste water must be correctly treated before final disposal, resulting in environmental protection as well as public health and socioeconomic problems.

It is a combination of sewage water from factory waste effluents, agricultural drainage and medical facilities; it is well known that household wastewater includes pathogens, suspended particles, and other organic and inorganic contaminants. These pollutants and impurities must be reduced to permissible levels for safe wastewater disposal in order to reduce environmental and health risks. As a result, removing organic pollutants and pathogens from wastewater is critical for its reuse in various activities. Sewage is water that flows after being utilised for home, industrial, manufacturing, and other reasons as waste. Water is the predominant ingredient of sewage, although additional constituents include organic waste and chemicals. Sewage discharge is one of the current issues confronting in many cities, and numerous efforts are being made to manage it like the Amrut Yojana. Sewage discharges are a major cause of water pollution, increasing the demand for oxygen and nutrient loading in bodies of water, fostering harmful algal blooms, and resulting in a destabilised aquatic environment. The problem is exacerbated in locations with rudimentary and inefficient wastewater treatment facilities. Traditional wastewater treatment systems used in developed countries are expensive to create, run, and maintain, particularly in decentralised communities. Research is being conducted to create treatment methods suitable for these decentralised communities.

The main goal of wastewater treatment is typically to allow the disposal of industrial and human effluents without endangering public health or doing unacceptable harm to the environment. Because aquaculture systems cannot use a similar method, the controller will need to be given more confidence in order to remediate waste water.

1.2 NECESSITY OF SAFE SANITATION

Water and sanitation may be viewed as the circumstances and activities pertaining to people's health, particularly the water supply and waste management systems. Such a work would obviously include other topics including solid wastes, industrial, and other unique or hazardous waste and storm water drainage. However, sewage is the most harmful of these contaminants. Unwanted situations, such as the creation of foul fumes, result from the breakdown of untreated sewage's organic content if it builds up and is allowed to become a septic. Untreated sewage also includes a variety of microorganisms that live in the human gastrointestinal tract. Aside from hazardous substances or substances that may be mutagenic or carcinogenic, sewage also includes nutrients that might promote the growth of aquatic vegetation. To safeguard the environment and public health, sewage must be removed from its sources of production immediately and without causing any annoyance, then it must be treated, reused, or dispersed into the environment sustainably.

1.3 INDIA'S CURRENT SCENARIO FOR URBAN SANITATION

The problem of sanitation is much worse in urban areas due to increasing congestion and density in cities. Indeed, the environmental and health implications of the very poor sanitary conditions are a major cause for concern. The study of Water and Sanitation Programme (WSP) of the World Bank observes that when mortality impact is excluded, the economic impact for the weaker section of the society accounting to 20 % of the households is the highest. The National Urban Sanitation Policy (NUSP) of 2008 has laid down the framework for addressing the challenges of city sanitation. The NUSP emphasizes the need for spreading awareness about sanitation through an integrated city-wide approach, assigning institutional responsibilities and due regard for demand and supply considerations, with special focus on the urban poor. As per the 2011 Census, 81.4% households have toilet facilities within their premises. This includes 70.9% households having water closets; 8.8% households having pit latrines; 1.7% households having other toilets (connected to open drains, night soil removed by human etc., which are unsafe). Out of the 70.9% households, 32.7% households have water closets connected to sewer system and 38.2% households are having water closets with septic tank. The remaining 18.6% households do not have toilet facilities within their 4 premises. This includes 6.0% households using public toilets and 12.6% households defecating in the open.

Sectoral demands for water are growing rapidly in India owing mainly to urbanization and it is estimated that by 2025, more than 50% of the country's population will live in cities and towns. Population increase, rising incomes, and industrial growth are also responsible for this dramatic shift. National Urban Sanitation Policy 2008 was the recent development in order to rapidly promote sanitation in urban areas of the country. India's Ministry of

Urban Development commissioned the survey as part of its National Urban Sanitation Policy in November 2008. In rural areas, local government institutions in charge of operating and maintaining the infrastructure are seen as weak and lack the financial resources to carry out their functions. In addition, no major city in India is known to have a continuous water supply and an estimated 72% of Indians still lack access to improved sanitation facilities.

1.4 BASIC PHILOSOPHY OF SEWAGE TREATMENT

The proper management of human excrement, including its safe containment, treatment, disposal, and related hygiene-related practises, is known as sanitation. It recognises that integral solutions must consider other aspects of environmental sanitation, such as solid waste management, generation of industrial and other specialised / hazardous wastes, drainage, as well as the management of drinking water supply, considering that this policy relates to the management of human excreta and associated public health and environmental impacts.

1.4.1 VISION

The goal of urban sanitation in India is to completely sanitise, make all Indian cities and towns healthy and livable, guarantee and sustain good public health and environmental outcomes for all of their citizens, with a special emphasis on providing hygienic and affordable sanitation facilities for the urban poor and women.

This policy's ultimate objective is to make Urban India's cities and towns community-driven, completely sanitised, healthy, and livable.

1.5 GOVERNMENT OF INDIA INITIATIVES

In 2014, India launched the Swachh Bharat Mission (SBM) – one of the world's largest drives for sanitation, which aimed to make the country open defecation free by 2019. Five years down the line, according to Government of India figures, 95 million toilets have been constructed, with access to household toilets shooting up from 37 percent in 2014 to 71 percent in 2018, and 520,000 villages being declared open-defecation free.

The Clean Ganga drive is a continuous process and the government has sanctioned 305 projects at an estimated cost of over Rs 28,600 crore under the plan. Minister for Jal Shakti

Shri Gajendra Singh Shekhawat said there has been improvement in the quality of water of the river Ganga.

Government of India launched the Namami Gange Programme with a total budgetary outlay of Rs 20,000 crore for 5-year period till 31 December 2020 to accomplish the twin objectives of effective abatement of pollution, conservation and rejuvenation of National River Ganga and its tributaries. "Due to various pollution abatement initiatives taken by the Government under the Namami Gange Programme, the water quality assessment of River Ganga in 2019 has shown improved water quality trends as compared to 2014," he said. Dissolved Oxygen levels have improved at 32 locations, Biological Oxygen Demand (BOD) levels and faecal coliforms have improved at 39 and 18 locations, respectively, he said. He said 379 MLD new sewage treatment capacity has been added in last three years of which 169 MLD has been achieved during last one year. The present sewage treatment capacity is 1954 MLD. Namami Gange Programme was launched in May 2015 as an integrated programme to ensure effective abatement of pollution and conservation of River Ganga by adopting a comprehensive river basin approach. Under Namami Gange Programme, diverse set of interventions for cleaning and rejuvenation of River Ganga have been taken up. These include pollution abatement activities including sewage, industrial effluent, solid waste etc., river front management, rural sanitation, afforestation, biodiversity conservation, public participation etc.

1.6 THE PROHABITATION AND ACTs

Environment (Protection) Amendment Rules, 2017 (Discharge Standard for Sewage Treatment Plants (STPs))

Effluent discharge standards (applicable to all mode of disposal) for Sewage Treatment Plants (STPs): In exercise of the powers conferred by sections 6 and 25 of the Environment (Protection) Act, 1986 (29 of 1986), the Central Government hereby makes the following rules further to amend the Environment (Protection) Rules, 1986, namely: - These rules may be called the Environment (Protection) Amendment Rules, 2017. They shall come into force on the date of their publication in the Official Gazette. These standards shall be applicable for 6 discharges into water bodies as well as for land disposal/applications. The standards for Faecal Coliform shall not apply in respect of use of treated effluent for industrial purposes. These Standards shall apply to all STPs to be commissioned on or after the 1st June, 2019 and the old/existing STPs shall achieve these standards within a period of five years from date of publication of this notification in the Official Gazette

PH	-6.5-9.0
Bio-Chemical Oxygen Demand (BOD)	-20 For Metro Cities
	30 Area/Regions other than Metro Cities
Total Suspended Solids (TSS)	< 50 For Metro Cities
	< 100 Area/Regions other than Metro Cities

Faecal Coliform (FC) (Most Probable Number per 100 millilitre, MPN/100mL <1000 anywhere in the country

*Metro Cities are Mumbai, Delhi, Kolkata, Chennai, Bengaluru, Hyderabad, Ahmedabad and Pune.

Note:

(i) All values in mg/l except for pH and Faecal Coliform.

(ii) These standards shall be applicable for discharge into water bodies as well as for land disposal/applications.

(iii) The standards for Faecal Coliform shall not apply in respect of use of treated effluent for industrial purposes.

(iv) These Standards shall apply to all STPs to be commissioned on or after the 1st June, 2019 and the old/existing STPs shall achieve these standards within a period of five years from date of publication of this notification in the Official Gazette.

(v) In case of discharge of treated effluent into sea, it shall be through proper marine outfall and the existing shore discharge shall be converted to marine outfalls, and in cases where the marine outfall provides a minimum initial dilution of 150 times at the point of discharge and a minimum dilution of 1500 times at a point 100 meters away from discharge point, then, the existing norms shall apply as specified in the general discharge standards.

(vi) Reuse/Recycling of treated effluent shall be encouraged and in cases where part of the treated effluent is reused and recycled involving possibility of human contact, standards as specified above shall apply.

(vii) Central Pollution Control Board/State Pollution Control Boards/Pollution Control Committees may issue more stringent norms taking account to local condition under section 5 of the Environment (Protection) Act, 1986".

1.7 OBJECTIVES OF WASTE WATER TREATMENT

The goal of municipal and industrial waste water treatment is to remove pollutants, toxicants, remove coarse particles, and kill pathogens in order to enhance the quality of discharged water and bring it up to the level that is allowed to be released into sources of water or for land use or for farming. Thus, water treatment seeks to reduce BOD, COD, eutrophication, etc. as well as avoid the bio-magnification of hazardous compounds in the food chain.

1.8 WHAT IS UASB TECHNOLOGY?

Anaerobic process is the foundation of UASB technology. The stabilisation of waste by anaerobic process does not require external oxygen. Anaerobic treatment of wastewater provides a variety of upsides over aerobic treatment, including reduced energy input of the system and decreased creation of surplus sludge per unit mass of stabilised organic matter. Biogas (a useful source of energy) is produced when waste organic materials degrade and decreased biological production reduces the need for nutrients. Anaerobic digestion, on the other hand, is a slow pace process that is reliant on the temperature and pH of waste water. The UASB process is also hampered by toxic contaminants. Because of the UASB process's cheap capital and operating & maintenance costs as well as low energy requirements, India places a specific emphasis on it. Anaerobic treatment stands out among the various choices for treating municipal and industrial effluents because it produces the least amount of sludge and produces energy in the form of methane. Research on the principles of anaerobic digestion has been ongoing for several decades, and the development of high-rate anaerobic systems has reduced the overall time of the digestion process. These high-rate digesters are quite modest in size and take up less room than other types. Tall reactors are being deployed in place of the flat, short reactors that were previously employed. Due to the retention of active granular settleable sludge in the reactor, loading rates for high-rate anaerobic digesters are rather high.

The UASB technique stands out among high-rate anaerobic digestion technologies due to its broad range of applications for all sorts of waste. The sole disadvantage of the procedure is that it starts off slowly in the absence of granular seed sludge.

The current study was carried out on a laboratory size UASB reactor treating residential waste water in the presence of digested sludge, and tests were carried out on the treated sludge retrieved every week from the pilot model to evaluate the effects of recirculation of the same waste water.

The following are the benefits of employing the UASB technique:

1) Lower energy consumption
2) Reduced formation of biological sludge
3) Requirement for fewer nutrients
4) Production of methane
5) Removal of offshore gas pollution
6) Quick reaction to substrate addition
7) No-feeding periods
8) Sludge dewatering plant methane can be transformed into biogas, but it can also be converted into hydrogen for use in direct fuel cells. As a result, another alternative for powering wastewater treatment facilities exists.

The UASB approach has the following disadvantages:

1) It is sensitive to the negative effects of low temperatures.
2) More prone to disturbances as a result of harmful chemicals.
3) Alkalinity may be required.
4) Biological N and P elimination is not a viable option.

CHAPTER

2. LITERATURE REVIEW

2.1 TECHNOLOGY FOR SEWERAGE AND SEWAGE TREATMENT

Environmental engineering includes sewerage and sewage treatment technologies. The fundamental concepts of engineering are used to address collection challenges. The basic concepts of biochemistry are applied to the treatment and environmental challenges of treated sewage disposal and reuse. The ultimate objective is to preserve public health while taking into account environmental, economic, social, and political considerations. To safeguard public health and the environment, it is vital to understand:

1. Concerning sewage constituents
2. The effects of these elements on the environment when sewage is discharged.
3. The transformation and long-term purpose of these constituents in treatment procedures.
4. Treatment procedures that can be employed to eliminate or change sewage elements.
5. Methods for making good use of or disposing of solids generated by treatment systems.

To provide an initial perspective in the field of sewerage and sewage treatment technology, a common terminology is first defined followed by:

1. A discussion of the issues that need to be addressed in the planning and design of sewerage management systems, and
2. The current status and new directions in sewerage and sewage treatment technology.

2.2 VARIATIONS IN SEWAGE FLOW

Variations in sewage flow and their effects on the design of various components of a sewerage scheme: -

The flows in these sanitary sewers fluctuate with water consumption on a seasonal, monthly, daily, and hourly basis, but they are sometimes delayed and less pronounced because of the storage space in sewers and the time required for the sewage to reach the point of gauging. In other words, the peak is flattened because it takes a significant amount of sewage to fill the sewers to the high flow point, and the high flow point and flow from various sections will reach the measuring site after varying durations of flow.

2.2.1 Peak Factor

Design flows for new towns may be computed using the design population and anticipated water consumption for residential, commercial, and industrial usage. The predicted population can be based on the final densities if a master plan with a land use pattern and zoning regulations is provided. Sewer flow changes hourly and throughout the year.

However, predicted peak flows are used for hydraulic design purposes. The contributing population, as shown in the table, determines the peak factor, or the ratio of maximum to average flows.

Table 1 Peak factor for contributory population

Contribution Population	Peak Factor
Up to 200000	3.00
200001 to 500000	2.50
500001 to 750000	2.25
Above 750001	2.00

The peak factor also depends upon the density of population, topography of the site, hours of water supply and hence, individual cases may be further analysed if required. The minimum flow may vary from 1/3 to 1/2 of average flow.
Source: CPHEEO, 1993

2.2.2 Minimum Velocity for Preventing Sedimentation

To ensure that deposition of suspended solid does not take place, self-cleansing velocity using Shield's formula is considered in the design of sewers.

$$V = \frac{1}{N}(R^{1/6}\sqrt{(K_s(G_s - 1)d_p)}$$

Where, n = Manning's coefficient

R = Hydraulic Mean Radius in m

K_S= Dimensionless constant with a value of about 0.04 to start motion of granular particles and about 0.8 for adequate self-cleaning of sewers

G_S = Specific gravity of particle

d_p = Particle size in mm

The above formula indicates that velocity required to transport material in sewers is mainly dependent on the particle size and specific gravity and slightly dependent on conduit shape and depth of flow. The specific gravity of grit is usually in the range of 2.4 to 2.65. Gravity sewers shall be designed for the velocities as in Table.

Table 2 Design velocities to be ensured in gravity sewers

No	Criteria	Value
1	Minimum velocity at initial peak flow	0.6 m/s
2	Minimum velocity at ultimate peak flow	0.8 m/s
3	Maximum velocity	3 m/s

Source: WPCF, ASCE, 1982

2.2.3 Maximum Velocity

The smooth interior surface of pipe gets scoured due to the continuous abrasion caused by the suspended solids present in sewage. This scouring and wear and tear of the pipe interior is much more pronounced at velocities higher than what can be tolerated by the pipe materials. This wear and tear of sewer will not only reduce their life spans but will also reduce their carrying capacities.

This limiting or non-scouring velocity will mainly depend upon the material of the sewer, and its value is given below for different commonly used sewer materials.

Table 3 Limiting velocity in m/sec for different sewer material

S.No.	Sewer Material	Limiting velocity in m/sec
1.	Vitrified tiles and glazed bricks	4.5-5.5
2.	Cast iron sewers	3.5-4.5
3.	Stone ware sewers	3.0-4.0
4.	Cement concrete sewers	2.5-3.0
5.	Ordinary brick-lined sewers	1.5-2.5
6.	Earthen channels	0.6-1.2

2.3 SEWAGE DECOMPOSITION

To guarantee the safe disposal of sewage, it is important to understand its traits and behaviour. This study will aid in identifying the level and kind of sewage treatment that is necessary in order to prevent sewage from being dumped in polluted areas.

The organic matter, which is decomposed by bacteria, under biological action, is called the biodegradable organic matter. Most of the organic matter present in sewage is biodegradable, and hence undergoes biological decomposition, which can be divided into two types, i.e.

2.3.1 Aerobic Decomposition, called aerobic decomposition; and
2.3.2 Anaerobic Decomposition, called putrefaction.

2.3.1 Aerobic Decomposition: The biodegradable organic matter will go through aerobic decomposition if air or oxygen is freely available to the wastewater in dissolved form. Both facultative bacteria and aerobic bacteria will be responsible for this process.

(i) Nitrogenous organic matter $\longrightarrow$ $NO_3^- + NH_3 \uparrow +$ Energy

(ii) Carbonaceous organic matter $\longrightarrow$ $CO_3^- + H_2O +$ Energy

(iii) Sulphurous organic matter $\longrightarrow$ $SO_4^{--} +$ Energy

2.3.2 Anaerobic Decomposition: If the wastewater does not have access to air or oxygen, putrefaction or anaerobic decomposition, will take place. The complex organic substance will subsequently be broken down into simpler organic molecules of nitrogen, carbon, and sulphur as anaerobic and facultative bacteria thrive.

(i) Nitrogenous organic matter $\xrightarrow[Anaerobics]{}$ $N_2 + NH_3 \uparrow +$ Organic Acids + Heat Energy

(ii) Carbonaceous organic matter $\longrightarrow$ $CO_2 \uparrow +$ Heat Energy

(iii) Sulphurous organic matter $\longrightarrow$ H_2S + Heat Energy

(iv) Organic acids $\longrightarrow$ $CH_4 \uparrow + CO_2 \uparrow +$ Heat Energy

2.4 BASIC PROCESSES OF BIOLOGICAL TREATMENT: -

Waste water contains lots of microorganisms so much so that each litre of sewage typically contains 5 to 50 billion bacteria. Only a small portion of these bacteria, known as pathogens, are dangerous to human health. However, a significant portion of the bacteria in sewage, known as non-pathogens, are crucial helpers in the process of breakdown and may thus be used in sewage treatment. Therefore, the main goal of sewage treatment is to create an environment where aerobic and anaerobic bacteria may stabilise the organic content in sewage by either aerobic or anaerobic breakdown.

Treatment units which work on oxidation alone (i.e., aerobic decomposition) are: Aeration tank, contact bed, intermitted sand filters, trickling filters, and oxidation ponds.

Treatment units which work on putrefaction alone (i.e., anaerobic decomposition) are: Septic tanks, Imhoff tanks, sludge digestion tanks, and anaerobic lagoons.

2.5 CHARACTERISTICS OF SEWAGE

Domestic sewage includes used water from the kitchen, bathroom and lavatory, among other places. The daily per capita usage of water, the quality of water supply, and the kind, condition, and size of the sewerage system, as well as people's behaviours, all contribute to variances in the characteristics of housing sewage. Municipal sewage, which includes both household and industrial effluent, might vary based on the kind of industry and industrial location. Here are some of the most essential aspects of sewage.

The quality of sewage can be checked and analysed by studying and testing its physical, chemical and bacteriological (biological) characteristics, as explained below:

2.5.1 Temperature

The temperature of sewage may be used to determine the solubility of oxygen, which impacts the transfer capacity of aeration equipment in aerobic systems and the rate of biological activity. Extremely low temperatures have a negative impact on the efficiency of biological treatment methods, as well as sedimentation efficiency. In general, the temperature of raw sewage in India is recorded to be between 15° and 35°C at various locations and seasons.

2.5.2 Colour and Odour

Depending on the concentration, fresh residential sewage seems somewhat soapy and hazy. Due to microbial activity, sewage turns stale with time, darkening in colour and emitting a strong odour.

2.5.3 Solids

Though sewage includes less than 0.5 percent particles and the rest is water, the annoyance produced by the solids cannot be neglected, since these solids are highly degradable and hence require adequate disposal. Sewage solids may be divided into three types: dissolved solids, suspended solids, and volatile suspended solids. Knowledge of the volatile or organic fraction of a material that decomposes is required because it comprises the burden on biological treatment units or the oxygen resources of a stream when sewage is disposed of through dilution. The assessment of organic and inorganic suspended particles provides an overall view of the load on the sedimentation and grit removal system during sewage treatment. When sewage is utilised for irrigation purposes or any other purpose, the dissolved inorganic part must be considered.

2.5.4 pH

The concentration of hydrogen ions, represented as pH, is an important parameter in the operation of biological systems. The pH of fresh sewage is somewhat higher than that of community-based water. However, organic matter decomposition may reduce the pH, and the presence of industrial effluent may cause significant oscillations. The pH of untreated sewage is typically between 5.5 and 8.0.

2.5.5 Nitrogen and Phosphorus

Proteins, amines, amino acids, and urea are the most common nitrogen compounds found in home sewage. The bacterial breakdown of these organic elements leads in ammonia nitrogen in sewage. Nitrogen is a fundamental component of biological protoplasm; hence its concentration is critical for the effective operation of biological treatment systems and land disposal. In general, home sewage includes enough nitrogen to meet the demands of biological treatment. If adequate nitrogen is not present in industrial effluent, it must be provided outside. The nitrogen level of untreated sewage is often found to be in the 20 to 50 mg/L range when tested as TKN.

Phosphorus contributes to residential sewage through phosphorus-containing food leftovers and their breakdown products. Increased usage of synthetic detergents significantly increases the phosphorus concentration of sewage. Phosphorus is also an important nutrient in biological activities. The phosphorus content in home sewage is often sufficient to sustain aerobic biological wastewater treatment. However, it will be a matter of concern if the treated effluent is reused. PO_4 concentrations in raw sewage are often found to be in the 5 to 10 mg/L range.

2.5.6 Chlorides

The concentration of chlorides in sewage exceeds the normal chloride level of drinking water. The chloride content in excess of the provided water can be utilised to determine the strength of the sewage. The daily contribution of chloride is around 8 g per person. Based on an average sewage flow of 150 LPCD, the chloride concentration of the sewage would be 50 mg/L greater than that of the supplied water. Any abnormal rise should be interpreted as the discharge of chloride-containing wastes or salty groundwater intrusion, the latter contributing to the sulphates and potentially leading to excessive hydrogen sulphide production.

2.5.7 Organic Matter

Organic chemicals found in sewage are of special importance in the field of environmental engineering. A wide range of bacteria interact with the organic material by utilising it as an energy or material source (which may be present in the sewage or the receiving water body). Metabolism refers to the use of organic material by microbes. Catabolism refers to the conversion of organic material by microorganisms to gain energy, whereas anabolism refers to the incorporation of organic material into cellular material.

To explain the metabolism of microorganisms and the oxidation of organic material, the concentration of organic matter in different forms must be quantified. Given the large diversity of organic compounds in sewage, determining these separately is completely impractical. As a result, a parameter must be employed to characterise a trait that all of these share. In practise, practically all organic compounds have two qualities that may be used: (1) organic compounds can be oxidised and (2) organic compounds contain organic carbon.

In environmental engineering there are two standard tests based on the oxidation of organic material:

1) Biochemical Oxygen Demand (BOD) and
2) Chemical Oxygen Demand (COD) tests

In both tests, the organic material concentration is measured during the test. The essential differences between the COD and the BOD tests are in the oxidant utilized and the operational conditions imposed during the test such as biochemical oxidation and chemical oxidation. The other method for measuring organic material is the development of the Total Organic Carbon (TOC) test as an alternative to quantify the concentration of the organic material.

2.5.7.1 Biochemical Oxygen Demand (BOD): The BOD of the sewage is the amount of oxygen required for the biochemical decomposition of biodegradable organic matter under aerobic conditions. The oxygen consumed in the process is related to the amount of decomposable organic matter. The general range of BOD observed for raw sewage is 100 to 400 mg/L. Values in the lower range are being common under average Indian cities.

2.5.7.2 Chemical Oxygen Demand (COD): The COD gives the measure of the oxygen required for chemical oxidation. It does not differentiate between biological oxidisable and non-oxidisable material. However, the ratio of the COD to BOD does not change significantly for particular waste and hence this test could be used conveniently for interpreting performance efficiencies of the treatment units. In general, the COD of raw sewage at various places is reported to be in the range 200 to 700 mg/L.

2.5.8 Toxic Metals and Compounds

Toxic heavy metals and compounds, such as chromium, copper, and cyanide, may make their way into municipal sewage via industrial emissions. The concentration of these chemicals is critical if the sewage is to be treated biologically or disposed of in a stream or on land. In general, these chemicals are not harmful in sanitary sewage; nevertheless, with industrial emissions, they may exceed dangerous limits in municipal wastewaters.

Table 4 General standards for discharge of wastewater pollutants

No	Characteristics	Standards			
		Inland Surface Water	Public Sewers, (A)	Land for Irrigation	Marine Coastal Areas
1	Colour and odour	(B)		(B)	(B)
2	SS	100	600	200	(C), (D)
3	Particle size of SS	(E)	-	-	(F), (G)
4	pH value	5.5 to 9.0			
5	Temperature	(H)	-	-	(H)
6	Oil and grease	10	20	10	10
7	Total residual chlorine	1.0	-	-	1.0
8	Ammoniacal nitrogen (as N)	50	50	-	50
9	Total Kjeldahl Nitrogen, (TKN) (as N)	100	-	-	100
10	Free ammonia (as NH_3)	5.0	-	-	5.0
11	Biochemical Oxygen Demand	30	350	100	100
12	Chemical Oxygen Demand	250	-	-	250
13	Arsenic (as As)	0.2			
14	Mercury (as Hg)	0.01	0.01	-	0.01
15	Lead (as Pb)	0.1	1.0	-	2.0
16	Cadmium (as Cd)	2.0	1.0	-	2.0
17	Hexavalent Chromium (as Cr 6+)	0.1	2.0	-	1.0
18	Total Chromium (as Cr)	2.0	2.0	-	2.0
19	Copper (as Cu)	3.0	3.0	-	3.0
20	Zinc (as Zn)	5.0	15.0	-	15.0
21	Selenium (as Se)	0.05	0.05	-	0.05
22	Nickel (as Ni)	3.0	3.0	-	5.0
23	Cyanide (as CN)	0.2	2.0	0.2	0.2
24	Fluoride (as F)	2.0	15.0	-	15.0
25	Dissolved phosphates (as P)	5.0	-	-	-
26	Sulphide (as S)	2.0	-	-	5.0
27	Phenolic compounds (as C_6H_5OH)	1.0	5.0	-	5.0
28	Radioactive materials				
	Alpha emitters, micro curie/L	10^{-7}	10^{-7}	10^{-8}	10^{-7}
	Beta emitters, micro curie/L	10^{-6}	10^{-6}	10^{-7}	10^{-6}
29	Bio-assay test	(I)			
30	Manganese (as Mn),	2.0	2.0	-	2.0
31	Iron (as Fe),	3.0	3.0	-	3.0
32	Vanadium (as V),	0.2	0.2	-	0.2
33	Nitrate Nitrogen (as N),	10.0	-	-	20.0
34.	Faecal Coliform, MPN/100 ml for discharge	onto land		into water	
		(J)	(K)	(J)	(K)
		1,000	10,000	1,000	10,000

2.6 WASTEWATER TREATMENT

Treatment and disposal of wastewater is necessary because wastewater collected from cities and towns must eventually be returned to receiving water or the land, this will promote environmental preservation and conservation. Once the minimum effluent quality has been characterised, the treatment's goal is to consistently meet the stated requirements for maximum permissible concentrations of particles (both suspended and dissolved), organic matter, nutrients, and pathogens. The design engineer's responsibility is to create a procedure that ensures the technical viability of the treatment process while also taking into account other issues like as construction and maintenance costs, the availability of building materials and equipment, and specialised personnel.

Primary treatment alone will not result in an acceptable residual organic material content in the effluent. Biological approaches are almost always utilised in treatment systems to do secondary treatment for organic material removal. Bacteria metabolise organic material in biological treatment systems. Depending on the final effluent quality requirement, tertiary treatment technologies and/or pathogen removal may be implemented.

The majority of wastewater treatment plants nowadays employ aerobic metabolism to remove organic materials. The most common aerobic methods include activated sludge, oxidation ditches, trickling filters, and aerated lagoons. Stabilization ponds use both the aerobic and anaerobic mechanisms. In recent years, as electricity costs have risen, so has the operating cost of aerobic processes, greater emphasis has been placed on the use of anaerobic treatment systems for the treatment of wastewater, including sewage. Recently, a high-rate anaerobic technique, such as an Upflow Anaerobic Sludge Blanket (UASB) reactor followed by an oxidation pond, has been employed for sewage treatment in a few sites.

2.6.1 Characterization Of Wastewater

After treatment, the effluent is ultimately disposed on land or into a body of water. In most cases, the treatment involves the removal of suspended or soluble organic debris and SS, which depletes the water body of DO. Even if the plant is built to completely remove this pollutant, the treatment will become unprofitable. Additionally, the current watercourses may absorb a small number of pollutants without negatively impacting the ecology. As a result, the majority of contaminants are eliminated in treatment facilities, leaving only the natural purification process for the remainder of the treatment. Therefore, before proceeding with the design of the treatment plant, it is essential to determine

1) The characteristics of the raw wastewater, and
2) The required degree of treatment i.e., the required characteristics of the treatment plant effluent.

Domestic wastewater characteristics fluctuate from industry to industry and from city to city, depending on people's standard of life and commercial and industrial activity in the city. In absence of any data for Indian cities, the per capita SS can be considered as 90 to 95 g/d and BOD as 40 to 45 g/d. The BOD associated with suspended solids is usually at a rate of 0.25 kg of BOD per kg of SS.

2.6.2 Characteristics of the Treatment plant effluent

The needed quality of treatment plant effluent is determined by the receiving water's quality criteria. The receiving water quality criteria are defined either by rules or by rigorous engineering study that takes into account natural purification or self-purification that happens in the receiving water. It can be regulated by Stream requirements that consider the assimilative capacity of the water body or discharge requirements that are administered generally under the authority's jurisdiction without considering the river water quality at individual locations. In India the effluent standards required for domestic sewage and industrial effluent is available on the Central Pollution Control Board (CPCB) website (http://cpcb.nic.in/GeneralStandards.pdf).

2.6.3 Sewage Treatment Flow Sheets

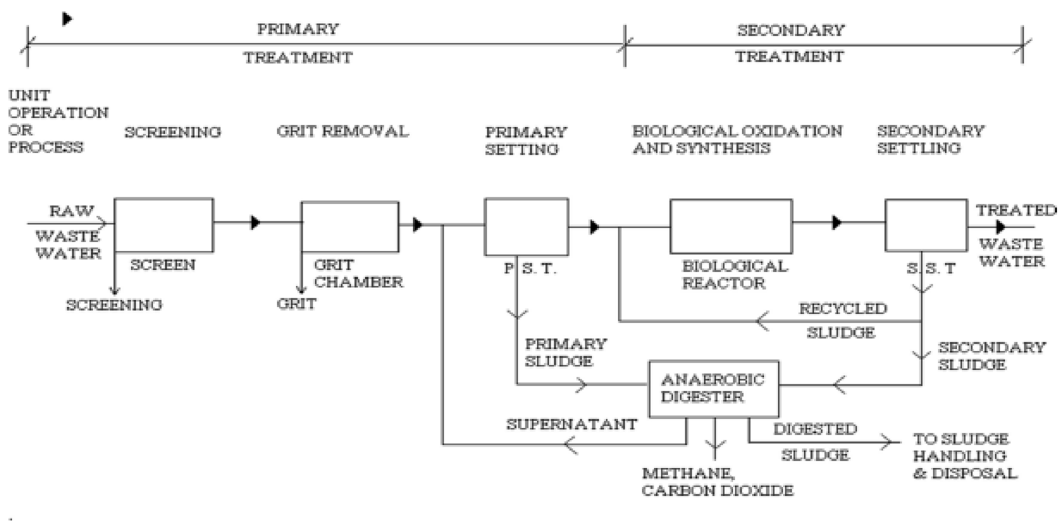

Figure 1 Process flow-sheet of conventional domestic sewage treatment plant

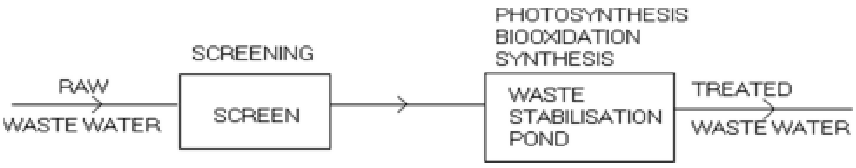

Figure 2 Process flow sheet employing waste stabilization pond

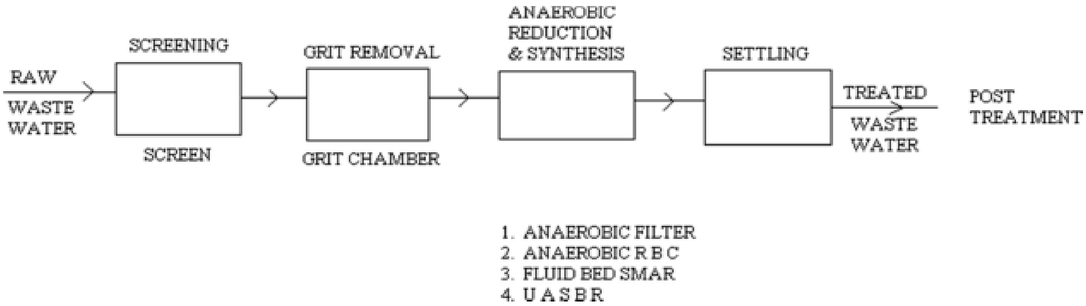

Figure 3 Process flow sheet employing anaerobic treatment system (CPHEEO, 2013)

Table 5 Water quality changes as freshwater becomes sewage and is gradually treated for reuse

			Water quality at different treatment steps			
Item	Fresh municipal water	Raw domestic sewage	After biological treatment	After coagulation and filtration	After softening & chlorination	After DS and DM
	(A)	(B)	(C)	(D)	(E)	(F)
pH	7.6-7.8	7.15-7.65	7.2-7.8	7.1-7.3	7.1-7.2	≈7.00
Total Hardness mg/l as $CaCO_3$	35-40	120-160	120-160	120-170	40 (a)	As arising
M.O. Alkalinity mg/l as $CaCO_3$	40-45	125-200	125-200	110-180 (b)	110-180	As arising
Chlorides mg/l, as Cl	15-20	60-130	60-120	60-130	60-130	As arising
Sulphates mg/l as SO_4	1.5-2.5	10-20	10-15	15-25	15 -25	As arising
Phosphates mg/l as PO_4	Traces - 0.1	6-16	3-10	0.2-0.5	0.2-0.5	Nil
Nitrates mg/l as NO_3	1.0-2.0	1.0-3.0	13-19	13-19	13-19	As arising
Silica mg/l as SiO_2	8-24	10-24	10-24	10-20	10-20	As arising
Dissolved solids mg/l	80-500	850–1,350.	850–1,350	800-1,350	600-1150	< 500
Suspended solids mg/l	5-10	350-450	15-30	Nil.	Nil.	Nil
Turbidity SiO_2, Units	5-10	Turbid	10-20	2.0-3.0	2.0-3.0	2.0–3.0
BOD, mg/l	0.1-1.5	200-250	6-10	1.0- 2.0	1.0-1.5	Nil
COD, mg/l	1.0-2.0	350-500	150-270	150-270	100-180	< 70
Bacteriological quality (as per coliform standards)	Safe	Unsafe	Unsafe	Safe	Safe	Safe
Specific conductance						700

2.7 CLASSIFICATION OF WASTE WATER TREATMENT

Various steps involved in treatment of waste water are as follows:

I. Preliminary Treatment

II. Primary Treatment

III. Secondary (or biological) treatment

IV. Complete final or tertiary treatment

2.7.1 PRELIMINARY TREATMENT

The only purpose of preliminary treatment is to separate the floating materials as well as heavy settleable inorganic particles. It also aids in the removal of oils, greases, and other contaminants from sewage. This treatment decreases the BOD content of wastewater by 15 to 30%. This treatment consists of the following processes discussed as given below:

2.7.1.1 Screening

In wastewater treatment, screens are the initial unit provided with the goal of removing large floating objects and substantial coarse solids. Waste water is sent through screens, grit chambers, and skimming tanks as part of this treatment process to remove debris, gross solids, grit, oil, and grease. The main intent of adding screens is to shield the pumps and other machinery from any harm caused by the sewage's floating matter.

A straight channel section should be provided a few meters ahead of the screen to ensure good distribution of flow across the screens. Hydraulically, flow velocity should not exceed 1.0 m/s in the channel, with 0.3 m/s considered good design. Head loss across the screen will depend on the degree of clogging. Clean bars and screens result in a head loss of less than 0.1 m. Provision should be made for a head loss of up to 0.3 m for manually cleaned or for manually operated, mechanically cleaned screens. Screens should be preferably be placed before the grit chambers.

Types of screens, their design and cleaning: - Screening devices in wastewater treatment are as

i) Manually cleaned bar rack – manually operated.

ii) Mechanically cleaned bar rack – mechanically operated.

Another type of classification is as

(a) Coarse screens – Bar screens consist of vertical bars equally spaced in a channel and inclined at an angle of 30 to 60 degree. Spacing between the bars is about 50mm or more. The approach velocity in the channel should not exceed 1.0 m/sec and should not be less than self-cleaning velocity. Velocity through screen should not exceed 0.3 m/sec at peak flow. Screens are either manually or mechanically operated. Every treatment plant should have both manually clean as well as mechanically clean screens.

(b) Medium screens – spacing between the bars is about 6mm to 40mm.

(c) Fine screens – Fine screens consists of woven wire mesh mounted on rotating drums and are partially submerged in waste water. Fine screens have perforations of 1.5 mm to 3.0 mm in size. Fine screens may be disc or drum type. Fine screens are commonly used in industrial waste water treatment.

Further, the velocity at low flows in the approach channel should not be less than 0.3 m/s to avoid deposition of solids. A straight channel before the screen is mandatory. Its length shall be a minimum of 5 times the width of the screen chamber. The slope of the hand cleaned screen should be in between 30 to 60° with horizontal.

The clear spacing between the bars may be in the range of 15 mm to 75 mm in case of mechanically cleaned bar screen. However, for the manually cleaned bar screen the clear spacing used is in the range 25 mm to 50 mm.

Head Loss on Screens

Head loss varies with the quantity and nature of screenings allowed to accumulate between cleanings. The head loss created by a clean screen may be calculated by considering the flow and the effective areas of the screen openings, the latter being the sum of the vertical projections of the openings.

Usually accepted practice is to provide loss of head of 0.15 m but the maximum loss of head with the clogged hand cleaned screen should not exceed 0.3 m. For mechanically cleaned screen, the head loss is specified by the manufacturer, and it can be between 150 to 600 mm.

The head loss through the cleaned or partially clogged flat bar screen can also be calculated using following formula:

$$h = 0.0729\,(V^2 - v^2)$$

Where, h = loss of head, m

V = velocity through the screen, m/sec

v = velocity before the screen, m/sec

The head loss through the fine screen can be calculated as:

$$h = \frac{1}{(2g.C_d)} * \left(\frac{Q}{A}\right)^2$$

Where, g = gravity acceleration (m/sec^2);

C_d is coefficient of discharge = 0.6 for clean rack;

Q is discharge through screen (m^3/sec); and

A is effective open submerged area (m^2).

Generally, it has been found that for all screenings from sanitary sewage vary from 0.0015 to 0.015 m^3/million litres with screen sizes having clear opening of 100 mm and 25 mm respectively.

Disposal of Screenings

In order to grind the screening and return it to the wastewater, grinders or disintegrator pumps might be used. Screenings may be disposed off at a sanitary landfill together with other municipal solid trash. The screens from major sewage treatment plants can be burned. Screenings from small wastewater treatment facilities might be buried on the plant's property as disposal.

2.7.1.2 Grit Removal

The grit chamber is the second unit operation utilised in wastewater preliminary treatment, and the goal is to remove suspended inorganic particles such as sandy and gritty matter from the wastewater. Grit chambers are of three major types as follows:

i. Velocity controlled V shaped long grit channels

ii. Square shaped chambers with entry and exit on opposite sides and mild hopper

iii. Vortex type cone and the centrifugal action plummets the grit to the bottom

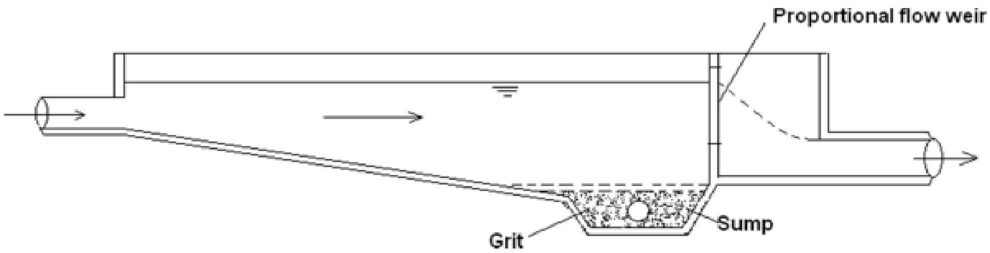

Figure 4 Horizontal flow grit chamber

Grits are inorganic solids with a specific gravity of 2.65, such as clay or sand. Grits are abrasive in nature and, if not removed, create wear and tear on mechanical equipment. They took up significant space in clarifiers and biological treatment systems as well. They should be separated from the organic suspended solids. Objective of Grit removal is to remove only inorganic particles while allowing organic solids to pass through. It is critical to maintain a consistent horizontal velocity of 0.3 m/s to allow grit removal.

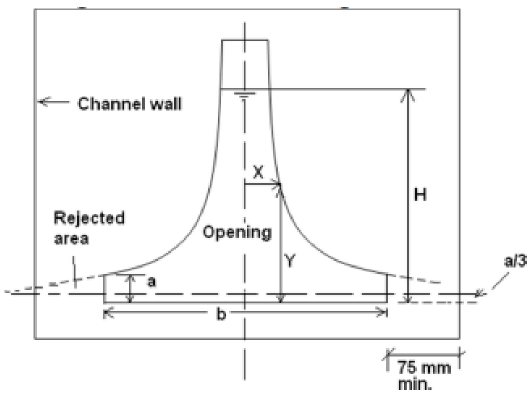

Figure 5 Proportional Weir

Constant horizontal velocity in a channel can be maintained by providing a parabolic section of the channel or a rectangular channel with a proportion weir put at the channel's exit. Grit chambers are hydraulically engineered to remove discrete particles with diameters of 0.2 mm and specific gravity of 2.65 by type I settling.

The tendency in larger treatment facilities is towards aerated grit chambers. Turbulence caused by compressed air injection keeps lighter organic material suspended while heavy grit settles to the bottom. Because roll velocity, rather than horizontal velocity, separates nontarget organics from grit, artificial control of horizontal velocity is not required. Air quantity adjustment gives settling control. The design of aerated grit chambers depends upon detention time. Aerated grit chambers may also be beneficial for other purposes. If the sewage is anaerobic when it arrives at the plant, aeration serves to remove toxic gases from the liquid and rapidly return it to an aerobic state, allowing for better treatment. When an aerated grit chamber is used, the aeration period is usually extended from 15 to 20 min.

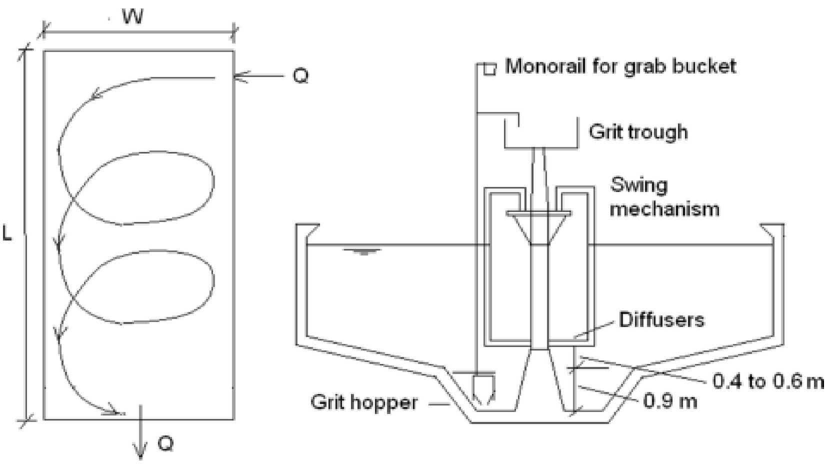

Figure 6 Aerated grit chamber (first figure showing the helical flow pattern of the wastewater in grit chamber and second showing cross section of grit chamber)

In aerated grit chamber –

Transverse velocity at surface 0.6-0.8 m/s

Depth-to-width ratio 1.5:1 to 2:1

Air supply 4.6-7.7 L/m/s of length 0.3-0.4 m^3 /min

Detention time 3-5 min (peak)

Quantity of grit 7.5-75 mL/m^3

Quantity of scum (skimming) 7.5-45 mL/m^3

2.7.2 PRIMARY TREATMENT

Primary treatment consists in removing large suspended organic solids. This is usually accomplished by sedimentation in settling basins. The liquid effluent from primary treatment, often contains a large amount of suspended organic, and has a high BOD (about 60% of the original).

Sedimentation Tank: Sedimentation tanks are typically used to remove part of the suspended organic solids that are too heavy to be removed as floating matter and too light to be removed by grit chambers (intended to extract only the heavy inorganic solids of size greater than 0.2 mm, and specific gravity 2.65). Thus, part of the organic matter that comes from the sewage effluent leaving the grit chambers will be removed by the sedimentation tanks. A comprehensive sewage treatment process involves two stages of sedimentation: the primary sedimentation, which takes place before the biological treatment, and the secondary sedimentation, which takes place after the biological treatment.

Primary Sedimentation: Primary sedimentation is a unit operation designed to concentrate and remove suspended organic solids from the wastewater. When primary treatment was considered sufficient as the total treatment, primary settling was the most important operation in the plant.

Most of the suspended solid in wastewater are in "sticky" in nature and flocculate naturally. Primary settling operations proceed essentially as type-2 settling without the addition of chemical coagulants and mechanical mixing and flocculation operations. The organic material is slightly heavier than water and settles slowly, usually in the range of from 1.0 to 2.5 m/h. Lighter materials, primarily oils and grease, float to the surface and must be skimmed off.

Either circular tanks or long-rectangular tanks are used for primary sedimentation. In rectangular tanks, scum is removed by allowing the sludge scrapers to pierce the surface as they make their way back to the effluent end of the tank. The removal of floating debris usually involves a scum weir or a transverse scum scraper. Floating material is sent to a collecting site some distance beyond the effluent weir. A skimmer arm is connected to the sludge scraper motor mechanisms in circular tanks. The scum is removed by wiping it up an inclined apron and into a scum trough. Between the effluent weir and the scum removal facilities in both situations, there has to be a scum baffle. In most cases, screens, unwashed grit, or digested sludge are used to dispose off separated scum.

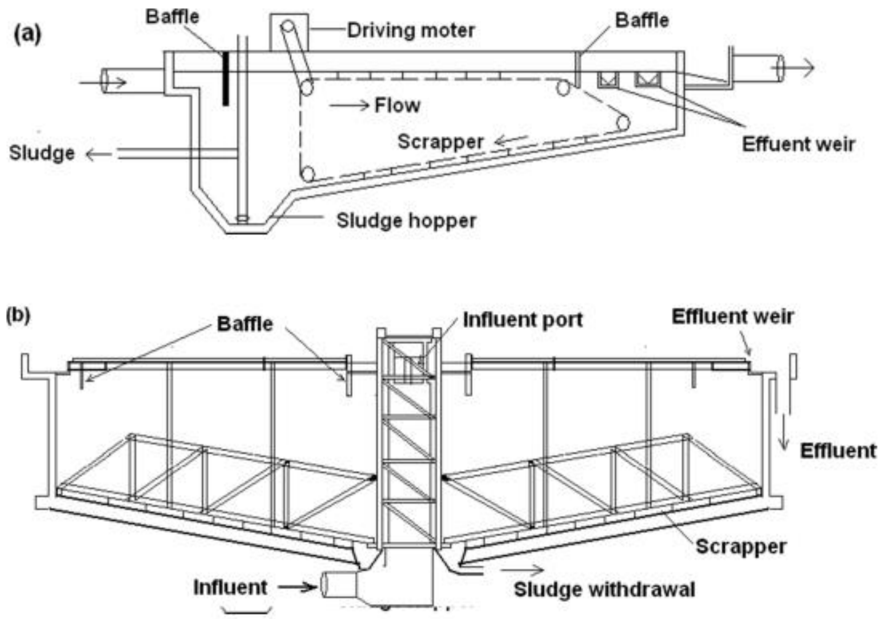

Figure 7 (a) Rectangular and (b) Circular primary sedimentation tank

Before anaerobic conditions arise, sludge in the primary sedimentation tank needs to be cleared. Gas bubbles will be formed if the sludge starts to break down anaerobically. These bubbles will stick to the solid particles and raise them towards the surface. After the sludge settles, it should be removed within 30 to 1 hour. A well-maintained system's sedimentation tank may remove 50 to 60 percent of the suspended particles and 30 to 35 percent of BOD from the sewage.

2.7.3 SECONDARY TREATMENT

Secondary treatment is the further treatment of effluent from the primary sedimentation tank. This is often performed by the biological breakdown of organic materials, which can occur under aerobic or anaerobic circumstances. Bacteria in these biological units will breakdown the fine organic matter. resulting in clearer effluent. The treatment reactors, in which the organic matter is decomposed (oxidized) by aerobic bacteria, are known as aerobic biological units; and may consist of

- Filters (Intermediate sand filters as well as trickling filters);
- Aeration tanks, with the feed of recycled activated sludge;
- Oxidation pond and Aerated lagoons.

The treatment reactors, in which the organic matter is destroyed and stabilized by anaerobic bacteria, are known as anaerobic biological units and may consist of

- Anaerobic lagoons
- Septic tanks
- Imhoff tanks, etc.

2.7.3.1 AEROBIC DEGRADATION OF SEWAGE

The main sedimentation tank effluent comprises 40 to 50 percent of the unstable organic materials initially found in sewage. This colloidal organic matter that passes through the primary clarifiers without settling must be removed by secondary or biological treatment, which is accomplished by changing the character of the organic matter and thus converting into stable forms (such as nitrates, sulphates, and so on) through oxidation or nitrification. The character of organic matter in sewage may be changed by different methods, which broadly classified into:

(i) Filtration
(ii) Activated sludge process.

Various types of filters that are commonly employed either singly or in combinations for giving secondary treatment of sewage are as:

i) Contact beds
ii) Intermittent sand filters
iii) Trickling filters
iv) Miscellaneous type of filters

2.7.3.1.1 Types Of Biological Reactors

Depending upon availability of oxygen or other terminal electron acceptor the biological reactors are classified as aerobic, anaerobic, anoxic or facultative process. Depending on how the bacteria are growing in the reactors they can be classified as

(a) **Suspended Growth Process**: where bacteria are grown in suspension in the reactor without providing any media support such as activated sludge process, and

(b) **Attached Growth Process**: where microorganism growth occurs as a biofilm formed on the media surface provided in the reactor such as trickling filters. This media could be made from rocks or synthetic plastic media offering very high surface area per unit volume. The media could be stationary in the reactor, as in trickling filter, which is called as fixed film reactor or it could be moving media as used in moving bed bioreactor (MBBR). Hybrid reactors are becoming popular these days which employ both suspended growth as well as attached growth in the reactor to improve biomass retention and substrate removal kinetics such as submerged aerobic filters (SAF).

2.7.3.1.2 Activated Sludge Process

Under aerobic circumstances, conventional biological treatment of wastewater comprises the activated sludge process (ASP) and the Trickling Filter. In 1914, the ASP was invented in England. The activated sludge process consists of an aeration tank in which organic matter is stabilised by the action of bacteria under aeration and a secondary sedimentation tank (SST) in which the biological cell mass is separated from the aeration tank effluent and the settle sludge is recycled partly to the aeration tank and the remainder is wasted. Recycling is required for the activated sludge process. Diffused or mechanical aeration is utilised to obtain the aeration conditions.

Diffusers are installed at the bottom of the tank, while mechanical aerators are installed at the surface of water, either floating or on fixed support. In the event of diffused air aeration, settled raw wastewater and returning sludge enter the tank head and traverse the tank following the spiral flow pattern, or get totally mixed in the case of a completely mixed reactor. In the case of a plug flow aeration tank, the air supply can be tapered along its length to match the amount of oxygen demand. In the settling tank, the effluent settles and the sludge is returned at the desired rate.

2.7.3.1.2.1 Aeration in ASP

Aeration units can be classified as:

1) Diffused Air Units
2) Mechanical Aeration Units
3) Combined Mechanical and diffused air units.

Diffused Air Aeration

Compressed air is blown around via diffusers in diffused air aeration. The tanks of these devices are typically small rectangular tubes. Air diffusers are installed at the tank's bottom.

To remove particles from the air before it passes through the diffusers, it must first travel through an air filter. Air compressors are used to keep the needed pressure.

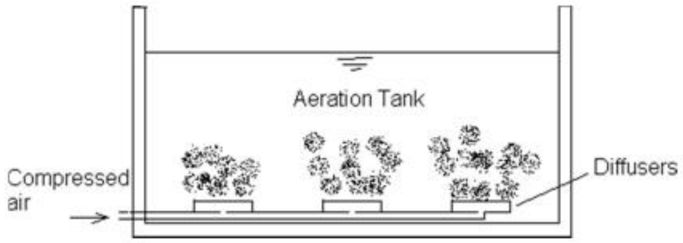

Figure 8 Typical air diffusers arrangement

Porous dome air diffusers with a diameter of 10 to 20 cm are commonly utilised. These are directly attached to the top of the C.I. main pipes, which are installed at the bottom of the aeration tanks. These are low in both initial and ongoing costs.

Mechanical Aeration Unit

The primary goal of mechanical aeration is to bring each new surface of wastewater into contact with air. Only 5 to 12% of the entire compressed air volume is used for oxidation in diffuse aeration, with the remainder used for mixing. Surface aerators of both the fixed or floating type can be utilised for this purpose. The rectangular aeration tanks are separated into square sections, each with its own mixer. When the electric motors start, the impellers suck the sewage from the centre, with or without tube support, and spray it over the surface of the wastewater in the form of a thin spray. When wastewater is sprayed into the air, more surface area of the wastewater comes into contact with the air, causing aeration to occur at a faster pace. The aeration tank handling sewage normally has a detention time of 5 to 8 hours. The capacity of the aeration tank should be calculated keeping the volume of the return sludge in mind.

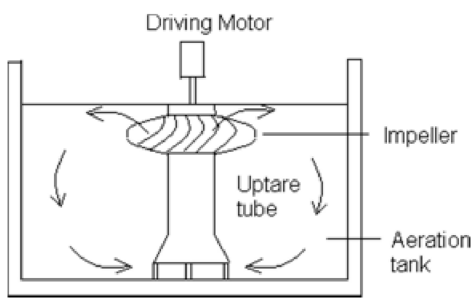

Figure 9 Typical arrangement of the surface aerator supported on conical bottom tube

2.7.3.1.2.2 Types of Activated Sludge Process

Conventional Aeration

In conventional ASP, the flow model in aeration tank is plug flow type. Both the influent wastewater and recycled sludge enter at the head of the tank and are aerated for about 5 to 6 hours for sewage treatment. The influent and recycled sludge are mixed by the action of the diffusers or mechanical aerators. Rate of aeration is constant throughout the length of the tank. During the aeration period the adsorption, flocculation and oxidation of organic matter takes place. The F/M ratio of 0.2 to 0.4 kg BOD/kg VSS/d and volumetric loading rate of 0.3 to 0.6 kg BOD/m^3/d is used for designing this type of ASP. Lower mixed liquor suspended solids (MLSS) concentration is maintained in the aeration tank of the order of 1500 to 3000 mg/L and mean cell residence time of 5 to 15 days is maintained. The hydraulic retention time (HRT) of 4 to 8 h is required for sewage treatment. Higher HRT may be required for treatment of industrial wastewater having higher BOD concentration. The sludge recirculation ratio is generally in the range of 0.25 to 0.5.

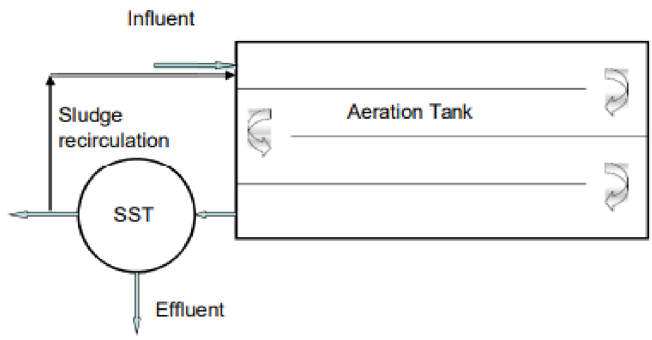

Figure 10 Conventional activated sludge process

Tapered Aeration

In plug flow type aeration tank, BOD load is maximum at the inlet and it reduces as wastewater moves towards the effluent end. Hence, accordingly in tapered aeration maximum air is applied at the beginning and it is reduced in steps towards end, hence it is called as tapered aeration. By tapered aeration the efficiency of the aeration unit will be increased and it will also result in overall economy. The F/M ratio and volumetric loading rate of 0.2 to 0.4 kg BOD/kg VSS/d and 0.3 to 0.6 kg BOD/m^3/d, respectively, are adopted in design. Other design recommendations are mean cell residence time of 5 to 15 days, MLSS of 1500 to 3000 mg/L, HRT of 4 to 8 h and sludge recirculation ratio of 0.25 to 0.5. Although, the design loading rates are similar to conventional ASP, tapered aeration gives better performance.

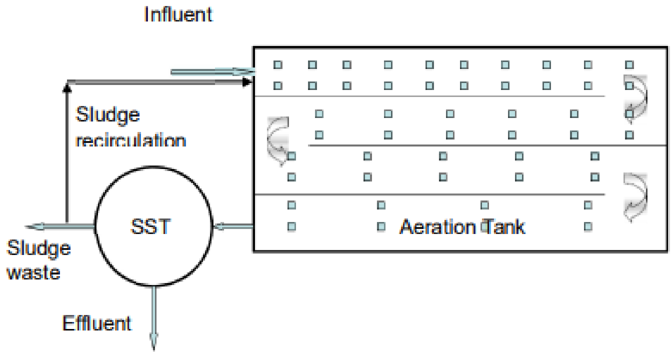

Figure 11 Tapered aeration activated sludge process

Step aeration

If the sewage is added at more than one point along the aeration channel, the process is called as step aeration. This will reduce the load on returned sludge. The aeration is uniform throughout the tank. The F/M ratio and volumetric loading rate of 0.2 to 0.4 kg BOD/kg VSS/d and 0.6 to 1.0 kg BOD/m^3/d, respectively, are adopted in design. Other design recommendations are mean cell residence time of 5 to 15 days, MLSS of 2000 to 3500 mg/L, HRT of 3 to 5 h and sludge recirculation ratio of 0.25 to 0.75. In step aeration the design loading rates are slightly higher than conventional ASP. Because of reduction of organic load on the return sludge it gives better performance.

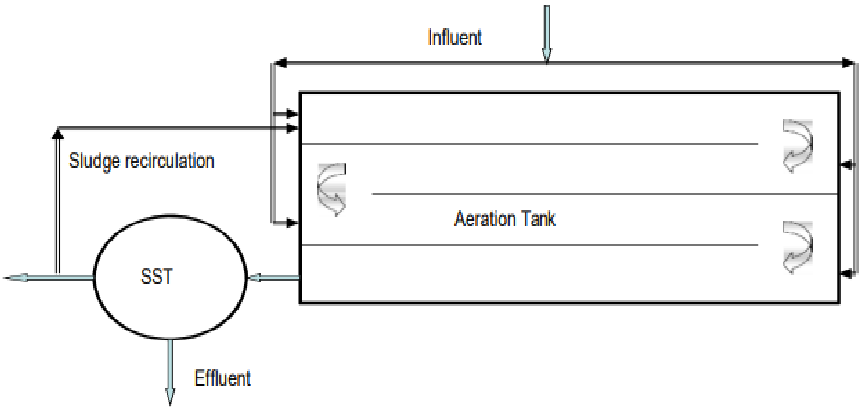

Figure 12 Step aeration activated sludge process

Completely mixed

In this type of aeration tank completely mixed flow regime is used. The wastewater is distributed along with return sludge uniformly from one side of the tank and effluent is collected at other end of the tank. The F/M ratio of 0.2 to 0.6 kg BOD/kg VSS/d and volumetric loading of 0.8 to 2.0 kg BOD/m^3/d is used for designing this type of ASP. Higher mixed liquor suspended solids (MLSS) is maintained in the aeration tank of the order of 3000 to 6000 mg/L and mean cell residence time of 5 to 15 days is maintained. The hydraulic retention time (HRT) of 3 to 5 h is required for sewage treatment. Higher HRT may be required for treatment of industrial wastewater having higher BOD concentration. The sludge recirculation ratio is generally in the range of 0.25 to 1.0. This type of ASP has better capability to handle fluctuations in organic matter concentration and if for some time any toxic compound appears in the influent in slight concentration the performance will not be seriously affected. Due to this property, completely mixed ASP is being preferred in the industries where fluctuation in wastewater characteristics is common.

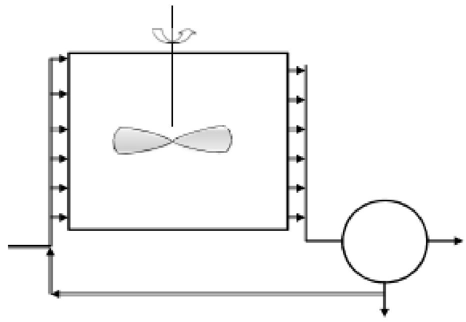

Figure 13 Complete mixed activated sludge process

Contact Stabilization

It is developed to take advantage of the absorptive properties of activated sludge. The BOD removal in ASP occurs in two phases, in the first phase absorption and second phase of oxidation. The absorptive phase requires 30 to 40 minutes, and during this phase most of the colloidal, finely divided suspended solids and dissolved organic matter get absorbed on the activated sludge. Oxidation of organic matter then occurs. In contact stabilization these two phases are separated out and they occur in two separate tanks (Figure). The settled wastewater is mixed with re-aerated activated sludge and aerated in the contact tank for 30 to 90 min. During this period the organic matter is absorbed on the sludge flocs. The sludge with absorbed organic matter is separated from the wastewater in the SST. A portion of the sludge is wasted to maintain requisite MLVSS concentration in the aeration tank. The return sludge is aerated before sending it to aeration tank for 3 to 6 h in sludge aeration tank, where the absorbed organic matter is oxidized to produce energy and new cells.

The aeration volume requirement in this case is approximately 50% of the conventional ASP. It is thus possible to enhance the capacity of the existing ASP by converting it to contact stabilization. Minor change in piping and aeration will be required in this case. Contact stabilization is effective for treatment of sewage; however, its use to the industrial wastewater may be limited when the organic matter present in the wastewater is mostly in the dissolved form. Existing treatment plant can be upgraded by changing the piping and providing partition in the aeration tank. This modification will enhance the capacity of the existing plant. This is effective for sewage treatment because of presence of organic matter in colloidal form in the sewage. Contact stabilization may not be that effective for the treatment of wastewater when the organic matter is present only in soluble form Excess sludge waste.

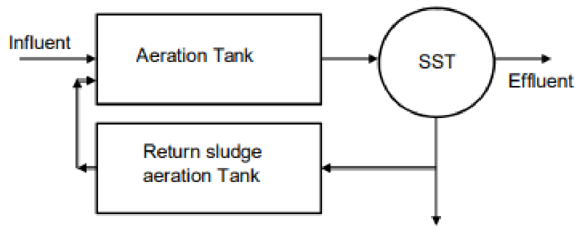

Figure 14 Contact stabilization activated sludge process

Extended Aeration

In extended aeration process, low organic loading rate (F/M) and long aeration time is used to operate the process at endogenous respiration phase of the growth curve. Since, the cells undergo endogenous respiration, the excess sludge generated in this process is low and the sludge can directly be applied on the sand drying beds where aerobic digestion and dewatering of the sludge occurs. The primary sedimentation can be eliminated when extended aeration process is used to simplify the operation of sludge handling. This type of activated sludge process is suitable for small capacity plant, such as package sewage treatment plant or industrial wastewater treatment plant of small capacity of less than 3000 m3/day. This process simplifies the sludge treatment and separate sludge thickening and digestion is not required. The aeration tank in this case is generally completely mixed type. Lower F/M ratio of 0.05 to 0.15 kg BOD/kg VSS/d and volumetric loading of 0.1 to 0.4 kg BOD/m^3/d is used for designing extended aeration ASP. Mixed liquor suspended solids (MLSS) concentration of the order of 3000 to 6000 mg/L and mean cell residence time of 20 to 30 days is maintained. Higher mean cell residence time is necessary to maintain endogenous growth phase of microorganisms. The hydraulic retention time (HRT) of 18 to 36 h is required. The sludge recirculation ratio is generally in the range of 0.75 to 1.5.

Sequencing batch reactor (SBR)

A sequencing batch reactor (SBR) is used in small package plants and also for centralized treatment of sewage. The SBR system consists of a single completely mixed reactor in which all the steps of the activated sludge process occur. The reactor basin is filled within a short duration and then aerated for a certain period of time. After the aeration cycle is complete, the cells are allowed to settle for a duration of 0.5 h and effluent is decanted from the top of the unit which takes about 0.5 h. Decanting of supernatant is carried out by either fixed or floating decanter mechanism. When the decanting cycle is complete, the reactor is again filled with raw sewage and the process is repeated. An idle step occurs between the decant and the fill phases. The time of idle step varies based on the influent flow rate and the operating strategy. During this phase, a small amount of activated sludge is wasted from the bottom of the SBR basin. A large equalization basin is required in this process, since the influent flow must be contained while the reactor is in the aerating cycle.

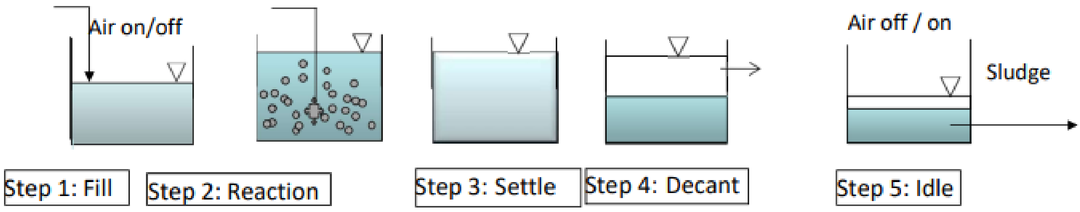

Figure 15 Operation cycles of sequencing batch reactor

This process is popular because entire process uses one reactor basin. In areas where there is a limited amount of space, treatment takes place in a single basin instead of multiple basins, allowing for a smaller footprint. In the effluent low total-suspended-solid values of less than 10 mg/L can be achieved consistently through the use of effective decanters that eliminate the need for a separate clarifier. The treatment cycle can be adjusted to undergo aerobic, anaerobic and anoxic conditions in order to achieve biological nutrient removal, including nitrification, denitrification and some phosphorus removal.

2.7.3.1.2.3 Limitations of ASP

For the treatment of wastewater with a high organic matter concentration, for example, if the resultant COD concentration in the aeration tank after dilution is in the thousands of mg/L, it will yield biomass equal to nearly half of the COD concentration. The overall biomass concentration in the system will be higher with the initial biomass concentration plus the created biomass concentration. This may make the operation of ASP problematic, such as evenly aerating the system at such high biomass concentrations, as well as settling and recirculation of the sludge. As a result, this method is not suggested for the first stage treatment of highly concentrated organic wastewaters.

2.7.3.1.3 Ponds System for Treatment of Wastewater

It is a relatively small body of water enclosed in an earthen basin that is exposed to the light and air. The pond has a longer retention duration ranging from a few days to weeks. The symbiotic connection between bacteria and algae results in wastewater cleaning. Ponds are characterised as aerobic, facultative, or anaerobic based on the type of biological activity that occurs inside them. Because they are less expensive to build and run in hotter climates than conventional treatment systems, they are referred to as low-cost wastewater treatment systems. However, they need a larger land area than conventional treatment systems.

2.7.3.1.3.1 Classification of Ponds

Aerobic Ponds: In addition to algae, an aerobic pond has a microbial community similar to ASP. The aerobic population produces CO_2, which the algae use for development. Algae, in turn, produce O_2, which aids in the pond's aerobic state. The shallow depth of an aerobic pond (0.15 to 0.45 m) is utilised for wastewater treatment to remove nitrogen through algal development. A depth of 0.5 to 1.2 metres can be employed for routine wastewater treatment. Solar radiation should penetrate to the pond's whole depth to enable photosynthesis and maintain the entire pond content aerobic. When shallow ponds (0.5 m deep) are utilised for tertiary wastewater treatment, they are relatively lightly laden, and these ponds are known as maturation ponds. During the day, these maturation ponds may release oxygen into the atmosphere.

Facultative stabilization Ponds: The majority of the ponds are of a facultative nature. This sort of pond has three zones. The upper zone is an aerobic zone where algal photosynthesis and aerobic biodegradation occur. Organic materials in wastewater and cells formed in the aerobic zone settle down and undergo anaerobic breakdown in the bottom zone. The intermediate zone is both aerobic and anaerobic in nature. Organic waste degradation in this zone is carried out by facultative bacteria. The existence of the upper aerobic zone eliminates the annoyance associated with the anaerobic response. Maintaining an aerobic state at the top layer is critical for the correct operation of the facultative stabilisation pond, and it is affected by sun radiation, wastewater properties, BOD loading, and temperature. The performance of these ponds is equivalent to that provided by conventional wastewater treatment systems.

Anaerobic pond: In anaerobic pond, the entire depth is under anaerobic condition except an extremely shallow top layer. For complete treatment, these ponds are often utilised in succession, followed by facultative or aerobic ponds. These ponds range in depth from 2.5 to 6 m. They are often used to treat high intensity industrial wastewaters, while occasionally they are also utilised to treat sludge and municipal wastewater. The anaerobic ponds retain a longer retention time of up to 50 days depending on the strength of the wastewater.

Nowadays, polyethylene sheets are used to cover anaerobic lagoons in order to recover biogas, eliminate odour issues, and reduce atmospheric emissions of greenhouse gases.

Fish pond: It can be part of maturation pond or altogether separate pond, in which fish are reared. Sometimes, fishes are also reared in the end compartment of primary pond.

Aquatic plant ponds: These are secondary ponds in which aquatic plants. e.g. hyacinths, duckweeds, etc. are allowed to grow either for their ability to remove heavy metals and other substances from wastewaters, or to give further treatment to wastewaters and produce new plant biomass. This recovered biomass can be used for biodiesel, bioethanol, combustible gas recovery as fuel or many other chemicals can be recovered using these plants as feed stock.

High-rate algal ponds: The high-rate algal pond (HRAP) may function as an efficient disinfection system that satisfies sustainability standards. The HRAP also has mechanisms for removing nutrients, notably those that remove phosphate, in addition to disinfection. These ponds are designed for maximal algal production rather than most effective water purification. The algae are collected to produce high-quality algal protein, but they are also used for other purposes. The ponds are 20–50 cm deep shallow lagoons with a 1–3 day retention time. By preserving a high algal population and utilising some sort of mechanical mixing, the entire pond is kept aerobic.

To avoid the formation of a sludge layer, mixing is typically done for short periods at night. To prevent an increase in pH in the surface water owing to photosynthesis, mixing may be necessary for brief periods of time during the day. Similar to a facultative pond, the pond is commissioned in the same manner, with the exception that continuous loading should be prohibited until an algal bloom has formed. Depending on solar radiation, the loading might range from 100 to 200 kg BOD each day on average throughout the year. Strong organic sewage blocks photosynthetic activity because of high ammonia levels, which causes the pond to turn anaerobic. High-rate algal ponds are made to encourage the symbiosis between aerobic bacteria and microalgae, with one using the other's main metabolic by-products.

Through photosynthesis, microalgae multiply rapidly and release oxygen from the water. The majority of the remaining soluble and biodegradable organic matter from the facultative pond may be quickly oxidised by bacteria thanks to this oxygen's instant availability. HRAPs have shorter hydraulic retention durations (HRTs) than facultative ponds and are shallower. The pH can rise over 9 when algae are growing quickly since that is when they are most active. There are too many hydroxyl ions after the carbonate and bicarbonate ions react, producing extra carbon dioxide for the algae. E. coli and likely the majority of harmful pathogenic microbes are completely destroyed by a pH above 9 for 24 hours.

Primary and secondary ponds: Raw or primary waste stabilisation ponds are defined as bodies of water that receive untreated wastewater. Secondary waste stabilisation ponds are those that accept primary treated or biologically treated wastewaters for further treatment. The maturation pond is the secondary pond that receives wastewater that has already undergone biological wastewater treatment, either from the ponds or another process like the UASB reactor or ASP. In these ponds, a detention period of 5 to 7 days is offered with the primary goal of producing the required levels of natural bacterial die-off. They frequently serve as an affordable alternative to chlorination in warm climates. They are not heavily loaded in terms of organic loading, and the amount of oxygen produced by photosynthesis could exceed the amount needed.

2.7.3.1.3.2 Typical Flow Chart of Pond Based Treatment Plant

The typical treatment flow sheets for different types of ponds in use are illustrated in the Figure. The ponds can be used in series or in parallel. Chlorination of the treated effluent is optional. The primary treatment after screen can be combined in the ponds along with secondary treatment. In all the flowcharts of the ponds in the Figure screens are provided ahead of the first pond.

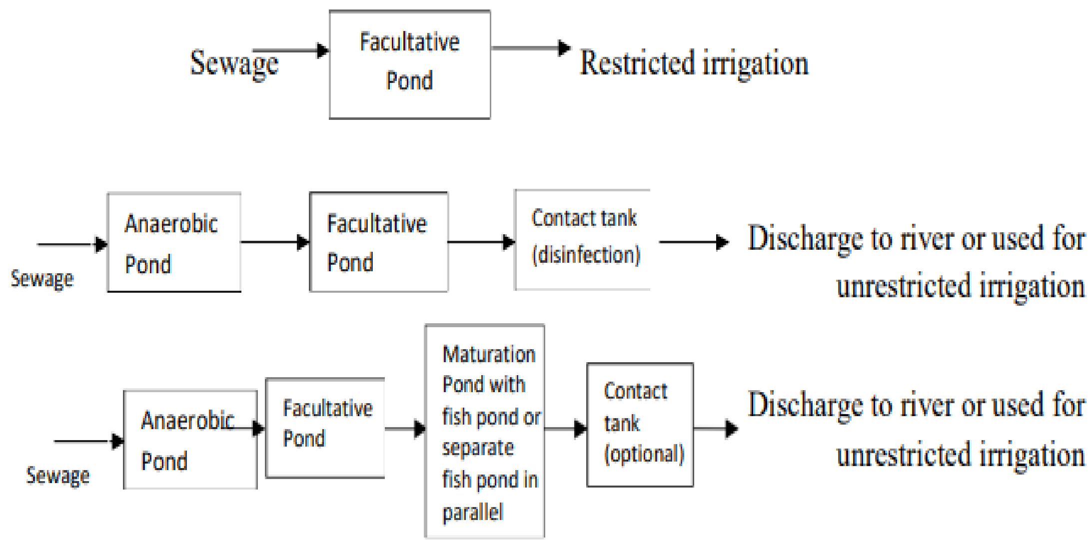

Figure 16 Flowcharts of the waste stabilization pond

2.7.3.1.3.3 Factors Affecting Pond Ecosystem

The principal abiotic components of ponds ecosystem are oxygen, carbon dioxide, water, light and nutrients; while the biotic components are algae, bacteria, protozoa, and variety of other organisms. Various factors affect the pond design, such as

- Wastewater characteristics and fluctuation,
- Environmental factors such as solar radiation, sky clearance, temperature, and their variation,
- Algal growth pattern and their diurnal and seasonal variation,
- Bacterial growth pattern and decay rates,
- Hydraulic transport pattern,
- Evaporation and seepage,
- Solids settlement, liquefaction, gasification, upward diffusion, sludge accumulation,
- Gas transfer at interface.

2.7.3.1.4 Aerated Lagoon

Aerated lagoons are one of the aerobic suspended growth processes. An aerated lagoon is a basin in which wastewater is treated either on a flow through basis or with solids recycle. Oxygen is usually supplied by means of surface aerators on floats or on fixed platforms or diffused air aeration units instead of photosynthetic oxygen yield as in case of oxidation pond. The action of the aerators and that of the rising air bubbles from the diffuser are used to keep the contents of the basin in suspension. They are constructed with depth varying from 2 to 5 m.

The contents of an aerobic lagoon are mixed completely. Depending on detention time, the effluent contains about 1/3 to 1/2 the value of the incoming BOD in the form of cell tissue. Before the effluent can be discharged, the solids must be removed by settling. If the solids are returned to the lagoon, there is no difference between this and modified ASP.

The mean cell retention time should be selected to assume, 1) that the suspended microorganisms will easily flocculate by sedimentation and 2) that the adequate safety factor is provided when compared to mean cell residence time of washout. The oxygen requirement is as per the activated sludge process. In general, the amount of oxygen required has been found to vary from 0.7 to 1.4 times the amount of BOD_5 removed.

Aerated lagoons have the advantages such as ease of operation and maintenance, equalization of wastewater, and a high capacity of heat dissipation when required. The disadvantages of aerated lagoons are large area requirement, difficulty in process modification, high effluent suspended solids concentration, and sensitivity of process efficiency to variation in ambient air temperature.

Aerobic lagoons: In aerobic lagoons, power levels are great enough to maintain all the solids in the lagoons in suspension and also to provide dissolved oxygen throughout the liquid volume. Aerobic lagoons are operated with high F/M ratio and short MCRT. These systems achieve little organic solids stabilization but convert the soluble organic material into cellular organic material. Based on the solid handling manner, the aerobic lagoons can be classified into (i) aerobic flow through with partial mixing, and (ii) Aerobic lagoon with solid recycle and nominal complete mixing.

Aerobic flow through with partial mixing: These types of aerobic lagoons operate with sufficient energy input to meet the oxygen requirement, but the energy input is insufficient to keep all the biomass in suspension. The HRT and SRT are the same in this type of lagoon. The effluent from this lagoon is settled in an external sedimentation facility to remove the solid prior to discharge.

Aerobic lagoons with solid recycle: These types of lagoons are same as the extended aeration activated sludge process with the exception that the aeration is carried out in an earthen basin instead of a reinforced concrete reactor basin and have longer HRT than the extended aeration process. The oxygen requirement in this type of lagoon is higher than the aerobic flow through lagoons to keep all the biomass in suspension. The analysis of this type of lagoons is same as the activated sludge process.

2.7.3.2 ANAEROBIC DEGRADATION OF ORGANIC MATTER

The factors that determine the removal efficiency of biodegradable organic matter are:

1. The nature and composition of the organic matter to be removed
2. Suitability of environmental factors
3. Sludge retention time in the reactor
4. The intensity of mixing, hence contact between bacterial biomass and organic matter.
5. Specific loading of organic matter with respect to bacterial sludge mass, and retention time.

Different steps are necessary for the anaerobic digestion of proteins, carbohydrates, and lipids. Four different phases can be distinguished in the overall conversion process of organic matter to biogas as

1) Hydrolysis, 2) Acidogenesis, 3) Acetogenesis, and 4) Methanogenesis

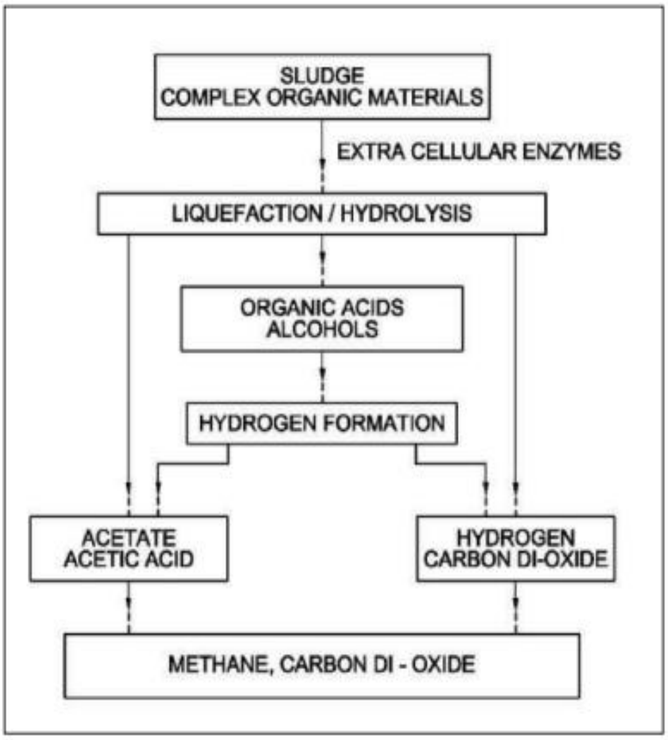

Figure 17 Anaerobic digestion of complex organic matter

2.7.3.2.1 Overview of Anaerobic Degradation Process

The anaerobic biological conversion of organic waste to methane is a complex process involving a number of microbial populations linked by their individual substrate and product specificities. The overall conversion process may be described to involve direct and indirect symbiotic association between different groups of bacteria. The product of one bacterium is often the substrate for others and hence, a balance between the bacterial numbers and the substrate concentrations must be maintained.

According to trophic requirements the bacteria involved can be conveniently divided into three groups as follows.

Hydrolytic bacteria-acidogens: These bacteria hydrolyse the substrate (macromolecule) into short-chain organic acids and other small molecules, which can be taken up and converted into soluble short-chain organic molecules, e.g., carbohydrates are converted into low-chain fatty acids, alcohols, hydrogen and carbon dioxide under anaerobic condition. Strict anaerobes are composed most part of this group of bacteria. The generation time of these bacteria is 2 to 3 hours. The distribution of final product depends on the species of acidogenic bacteria and on the environmental conditions such as pH and temperature.

Obligate Hydrogen Producing Acetogens (OHPA): This group converts compound formed in the first stage into acetic acid and hydrogen.

$$CH_3CH_2COOH + 2H_2O \longrightarrow CH_3COOH + CO_2 + 3H_2 \quad \Delta G_o = 76.1 \text{ KJ/mole}$$

Butyrate $$CH_3CH_2CH_2COOH + 2H_2O \longrightarrow 2CH_3COOH + 2H_2 \quad \Delta G_o = 48.1 \text{ KJ/mole}$$

From the viewpoint of the thermodynamics, a negative value of free energy change is necessary for any reaction to proceed without input of external energy. This theory apparently suggests that hydrogen producing acetogenic bacteria cannot obtain energy for growth from these reactions. However, the value of free energy change in the actual environment surrounding the bacteria, ΔG', is different from that of ΔGo' and depends on the concentrations of substrates and products.

Hydrogen utilizing methanogenic bacteria can serve such a thermodynamically favourable conditions for hydrogen producing acetogenic bacteria in anaerobic reactors, thus, the activity of hydrogen producing acetogenic bacteria depends on the existence of methanogenic bacteria. Hydrogen utilizing methanogens receive hydrogen as a substrate from hydrogen producing acetogenic bacteria. The interrelationship between these two groups of bacteria is called interspecies hydrogen transfer, which also exists between acidogenic and methanogenic bacteria. Acidogenic bacteria produce more hydrogen and acetate than propionate or lactate and obtain more energy under low hydrogen partial pressure which is kept by methanogenic bacteria. The interspecies hydrogen transfer is favourable but not essential for acidogenic bacteria, while, it is indispensable for hydrogen producing acetogenic bacteria.

Methanogenic bacteria - methanogens: These bacteria produce methane. The doubling time of these bacteria is 2 - 10 days. These are further divided into two groups as:

a) Hydrogen utilisers (lithotrophs)

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O \quad \text{convert ADP to ATP}$$

b) Acetic acid users (acetotrophs)

$$CH_3COOH \rightarrow CH_4 + CO_2 \quad \text{produce 0.25 mole of ATP}$$

The methane producing bacteria are strict anaerobes which are extremely sensitive to changes in temperature and pH. These bacteria are active in two temperature zones, namely, in themesophilic range (30°C – 35°C) and in the thermophilic range (50°C – 60°C). However, anaerobic processes have been operated at 15°C successfully when sufficient residence time for these bacteria was provided. The majority of methanogens in anaerobic wastewater treatment and natural anaerobic environment utilize hydrogen and single carbon compounds as substrates for methane production.

2.7.3.2.2 Factors Affecting Anaerobic Digestion

Development of anaerobic process technology is dependent on a better understanding of the factors that are associated with the stability of the biological processes involved. Process instability is usually indicated by a rapid increase in the concentration of volatile acids in the first stage of the reaction. Low pH with a concurrent reduction in methane gas production indicates the methanogenesis more susceptible to upset. Several environmental factors can affect anaerobic digestion such as specific growth rate, decay rate, gas production, substrate utilization, etc. The environmental factors of primary importance are discussed below.

2.7.3.2.2.1 pH, Acidity and Alkalinity

Methanogenic microorganisms are susceptible to the minute changes in the pH values. Optimum pH range of 6.6 – 7.6 is considered favourable for the methane producing bacteria, which cannot tolerate the fluctuations. The non-methanogenic bacteria do not exhibit such strong sensitivity for environmental conditions and are able to function in a range of pH form 5 – 8.5. The pH maintained inside the reactor, due to the process results from the interaction of the carbon dioxide-bicarbonate buffering system and volatile acids-ammonia formed by the process. It is necessary to prevent the accumulation of acids to a level, which may become inhibitory to the methanogenic bacteria. For this, it is important that there should be sufficient buffering capacity present in the reactor, which may prevent the reactor from souring. Although, the carbonates and bicarbonates of sodium and calcium are required to be added to the digesters to provide buffering action, lime (Calcium hydroxide) is most commonly used for this purpose. Only the unionized volatile acids in the concentration range of 30 - 60 mg/L are toxic.

2.7.3.2.2.2 Temperature

The higher the temperature, higher is the microbial activity until an optimum temperature is reached. A further increase of the temperature beyond its optimum value results in steep decrease in activity. Anaerobic process can take place over a wide range of temperatures (4° – 60°C). Once as effective temperature is established, small fluctuations can result in a process upset. Although most of the sludge digester are operated in the mesophilic range

(30° – 40°C), methanogenesis can occur at temperatures as low as 12° to 15°C. The optimum temperature for growth of anaerobic microorganisms is 35°C or greater.

2.7.3.2.2.3 Nutrients

Anaerobic wastewater treatment processes are often used for industrial waste with only minor amount of nutrients present. This might result in nutrient deficiency, unless additional nutrients are supplemented. Often the COD/N ratio and COD/N/P ratio is used to described the nutrient requirements. Optimum N/P ratio can be considered to be 7. The theoretical minimum COD/N–ratio is considered to be 350/7. A value around 400/7 is considered reasonable for high-rate anaerobic processes.

Other than nitrogen and phosphorous, trace metals also are essential for anaerobic processes. The presence of trace metals such as molybdenum, selenium, tungsten and nickel is probably necessary for the activity of several enzyme systems. When these trace elements are not present in the wastewater, addition of nickel, cobalt, and molybdenum can increase methane production and allow greater volumes of wastewater to be effectively treated by decreasing the reactor residence time.

2.7.3.2.3 Inhibitory Substances

Inhibition of the anaerobic digestion process can be mediated to varying degrees by toxic materials present in the system.

Volatile Acids Inhibition: Anaerobic reactor instability is generally evident by a marked and rapid increase in volatile fatty acids concentrations; this is frequently indicative of the failure of the methanogenic population due to other environmental disruptions such as shock loadings, nutrient depletion or infiltration of inhibitory substances. Acetate has been described as the least toxic of the volatile acids, while propionate has often been implicated as a major effecter of digester failure.

The inhibition by the volatile acids at acidic pH values can be attributed to the existence of unionized VFAs in significant quantities in the system. The undissociated nature of these acids allow them to penetrate the bacterial cell membrane more efficiently than their ionized counterparts, and once assimilated, induce an intracellular decrease in pH and hence a decrease in microbial metabolic rate. The resulting VFA concentration in the reactor should be maintained below 500 mg/L at any point of time and preferably below 200 mg/L for optimum performance.

Ammonia – Nitrogen Inhibition: Although ammonia is an important buffer in anaerobic processes, high ammonia concentration can be a major cause of operational failure. In reactor system that has not previously been acclimated to high ammonia loadings, shock loadings of high ammonia concentration generally caused rapid production of VFAs such that the buffering capacity of the system may not be able to compensate for the decrease in pH. Further depression of alkalinity and reduction of pH may result in reactor failure. Inhibition is indicated by a decrease in gas production and an increase in volatile acid formation.

Sulphide Inhibition: The sulphate and other oxidized compounds of sulphur are easily reduced to sulphide under the conditions prevalent in anaerobic digesters. Sulphur containing amino acids of protein can also undergo degradation to sulphide. These compounds are of significance when anaerobic treatment is considered for industrial processes which tend to produce large quantities of sulphides in their waste stream. These sulphides formed by the activity of reactor microorganisms may be soluble or insoluble, depending upon their associated cations. When the salts formed are insoluble, they have negligible effects on digestion. Iron addition, for example can suppress sulphide inhibition by removing S^{2-} ion from solution by precipitation.

All the heavy metals, with the exception of chromium, form insoluble sulphide salts and thus can be removed from solution by sulphide present in the system by precipitation. Free sulphide can also be eliminated as hydrogen sulphide by vigorous gas production.

Heavy Metals Inhibition: The most common agents of inhibition and failure of sewage sludge digesters are identified to be contaminating heavy metals. Heavy metals in the soluble state are in general regarded to be of more significance to reactor toxicity than are insoluble forms. Anaerobic digestion also reduces the valence states of some heavy metals. Both copper and iron may be reduced from the trivalent to divalent state. This reduces the quantity of the precipitating agent, such as sulphide, necessary for the removal of the metal ion from solution. The heavy metals can be removed from anaerobic systems by adsorption. The metals like copper, chromium, nickel, lead can induce toxicity in the reactor when present in higher concentration, and acceptable concentration in the wastewater to be treated differs from metal to metal.

2.7.3.2.4 Merits of Anaerobic Decomposition Process

It has been recognized that the anaerobic treatment is in many ways ideal for wastewater treatment and has several merits mentioned as below:

- A high degree of waste stabilization;
- A low production of excess biological sludge and this sludge can be directly dried on sludge drying bed without further treatment due to better dewatering ability;
- Low nutrient requirements, hence anaerobic treatment is attractive for the treatment of wastewater where external nutrient addition is required;
- No oxygen requirement, hence saving in power required for supply of oxygen in aerobic methods;
- Production of valuable by-product, methane gas;
- Organic loading on the system is not limited to oxygen supply hence higher loading rate as compared to aerobic processes can be applied;
- Less land required as compared to many aerobic processes;
- Non-feed conditions for few months do not affect adversely to the system and this makes it attractive option for seasonal industrial wastewater treatment.

Quantity of biological solids produced in the anaerobic systems per unit weight of organic material is much less than that in aerobic systems. This is a major advantage of the anaerobic process as the quantity of sludge for ultimate disposal is reduced. This is a result of conversion of volatile solids present, to the high energy level end products such as methane, carbon dioxide and water. Methane has a definite economic value as a fuel, and it is used as a source of energy for both heat and power in many installations. Another major advantage is the loading potential. Aerobic processes are restricted in maximum organic loading rate by the inability to transfer oxygen at the rate sufficient to satisfy the oxygen demand of the systems. Such limitation in organic loading rate is not there for anaerobic processes.

However, anaerobic treatment processes are not largely being implemented, because of many factors. Anaerobic microorganisms, especially methanogens have slow growth rate. At lower HRTs, the possibility of washout of biomass is more prominent due to higher upflow velocity. This makes it difficult to maintain the effective number of useful microorganisms in the system. To maintain the population of anaerobes, large reactor volume or higher HRTs with low upflow velocity is required. This may ultimately provide longer SRTs more than 40 days for high-rate systems. Thus, provision of larger reactor volume or higher HRTs ultimately leads to higher capital cost. Low synthesis / reaction rate hence, long start-up periods and difficulty in recovery from upset conditions are some of the notable disadvantages. Special attention is therefore required towards controlling the factors that affect process adversely; importantly among them being environmental factors such as, temperature, pH, and concentration of toxic substances. Hence, skill supervision is required for operating anaerobic reactor at optimal performance.

2.7.3.2.5 Anaerobic Waste Treatment Processes

Advantage of anaerobic waste treatment systems as means for recovery of non-conventional energy is increasingly being recognized worldwide. Anaerobic decomposition is a biologically mediated process, indigenous to nature, and capable of being simulated for treating wastes emanating from municipal, agricultural, and industrial activities. Anaerobic digestion as applied in treatment of sewage sludge and other organic wastes represents the controlled application of a process. The technological approaches to allow this condition of independent sludge retention time can be divided in to the following:

- Attachment of biomass on the media (filters, fluidized systems, and RBC configurations);
- Non-attached biomass concept as suspended growth process (sludge blanket reactors and contact process with sludge recycling).

Table 6 Basic types of reactors used in anaerobic process

Type of reactor	Synonyms	Abbreviations
Attached Biomass		
Fixed Bed	Fixed film, filter, submerged filter, /stationary fixed bed	SMAR (Submerged anaerobic reactor)/ ANFIL (upflow anaerobic filter), AUF (Anaerobic upflow filter), ADSR (Anaerobic downflow stationary bed reactor), AF (anaerobic filter), DSFF (downflow stationary fixed film reactor)
Moving Bed	Rotting discs, Rotating biological contactor,	AnRBC, RBC
Expanded bed	Anaerobic attached film expanded bed	AAFEB
Fluidized bed	Anaerobic Fluidized bed reactor, Carrier-assisted contact process	FBBR / IFCR (Immobilized fluidized cells reactor)/ CASBER (Carrier-assisted sludge bed reactor)
Non-attached Biomass		
Recycled flocks, Sludge blanket, Digester	Contact process, Upflow anaerobic sludge blanket reactor (UASB), Upflow sludge blanket (USB), Clarigester type	UASB, USB

The anaerobic contact process essentially involves two phases. A contact phase, where the raw waste is intimately mixed with a previously developed and available anaerobic sludge culture in the reactor; and a separation phase, where the active sludge particles are separated from the treated liquor and recycled to the contact unit. In this process, raw wastewater is mixed with recycled sludge solids and then digested in a digestion chamber. After digestion, the mixture is separated in a clarifier (or vacuum flotation unit), and the supernatant is discharged as effluent. Settled sludge is then recycled to seed the incoming waste.

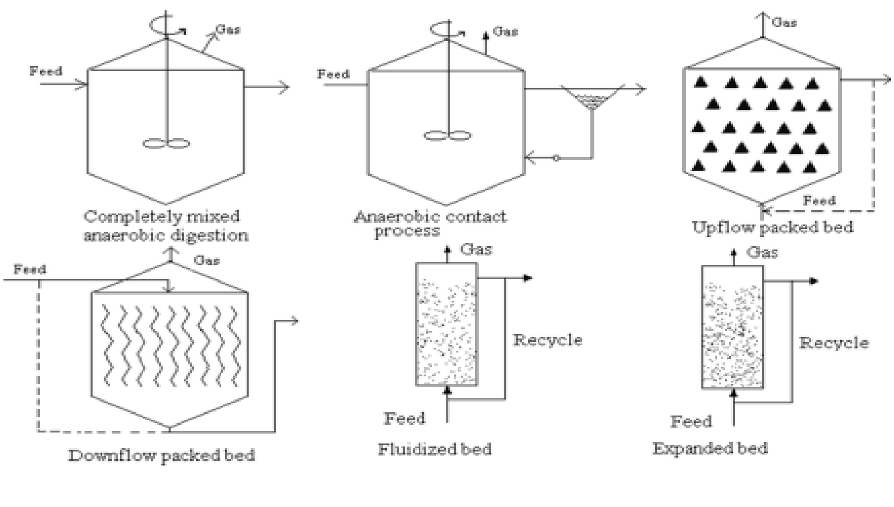

Figure 18 Typical reactor configurations used in anaerobic wastewater treatment

2.7.3.2.5.1 Anaerobic Filter Systems

The anaerobic filter (AF) and more recent variation of the filter process, the downflow stationary fixed film (DSFF) reactor, are packed with a fixed support media. In the DSFF reactor, the biomass is present as a biofilm attached to the support media. In the anaerobic filter, most of the biomass is present as suspended and/or entrapped biomass in the interstitial pore volume of the support media. The other major difference between these reactors is the direction of liquid flow through the packing. In the AF, the feed enters at bottom of the reactor. In the DSFF reactor, influent is applied in downward direction from the top of the rector. Both reactors can be used to treat either diluted or concentrated soluble wastewaters. Because of the relatively large clearance between the channels in the vertically oriented media used, the DSFF reactor is able to treat wastewater with relatively high suspended solids while the AF cannot.

Because the bacteria are retained on the media and not washed off in the effluent, mean cell residence times of the order of 100 days can be obtained. Large values of efficiency can be achieved with short hydraulic retention times, so the anaerobic filters can be used for the treatment of low-strength wastewater at ambient temperature. In the AF, most of the biological activity is due to the biomass in suspension (entrapped) rather than to the attached biofilm. The media with a high capacity to entrap and prevent washout of the biomass from the reactor is more important than the specific surface area (surface area to volume ratio) of the media. The biofilm thickness of 1 to 3 mm has been observed in fixed-bed reactor.

In the DSFF reactor systems, virtually all of the active biomass is attached to the support media. Different types of support media such as needle punched polyester (NPP) and red drain tile clay, PVC or glass can be used. For NPP, this attachment is probably associated with its surface roughness. The leaching of minerals from the clay could potentially stimulate bacterial activity and adhesion to this media support. The hydraulic retention time is generally kept in the range of 18 to 24 h, but lower HRT values can also give fairly good removal efficiency depending on the type of organic matter present in the wastewater.

2.7.3.2.5.2 Expanded Bed Process

In the expanded bed process, the wastewater to be treated is pumped upward through a bed of appropriate medium (e.g. sand, coal, expanded aggregate, plastic media) on which a biological growth has been developed. Effluent is recycled to dilute the incoming wastewater and to provide an adequate flow to maintain the bed in an expanded condition.

Biomass concentrations exceeding 15,000 to 40,000 mg/L can be developed. Since more biomass can be maintained, the expanded bed process can also be used for the treatment of low strength wastewater, such as municipal sewage, at very short hydraulic retention times. Organic loading in the range of 5 to 10 kg COD/m^3/d can be applied with COD removal efficiency of 80 to 85 %. The hydraulic retention time generally will be in the range of 5 to 10 hours.

2.7.3.2.5.3 Anaerobic Contact Process

The essential feature of the anaerobic contact process is that the washout of the active anaerobic bacterial mass from the reactor is controlled by a sludge separation and recycles system. The major problem in the practical application of the contact process has always been the separation (and concentration) of the sludge from the effluent solution. For this purpose, several methods have been used or were recommended for use, e.g. plain sedimentation, settling combined with chemical flocculation, with vacuum degasification, floatation and centrifugation. A basic idea underlying the contact process is that it is considered necessary to thoroughly mix the digester contents e.g., by gas recirculation, sludge recirculation, or continuous or intermittent mechanical agitation. This is generally used for concentrated wastewater treatment such as distillery wastewater.

2.7.3.2.5.4 Upflow Anaerobic Sludge Blanket (UASB) Reactor

Anaerobic process is the foundation of UASB technology. The stabilisation of waste by anaerobic processes does not require external oxygen. Anaerobic treatment of wastewater

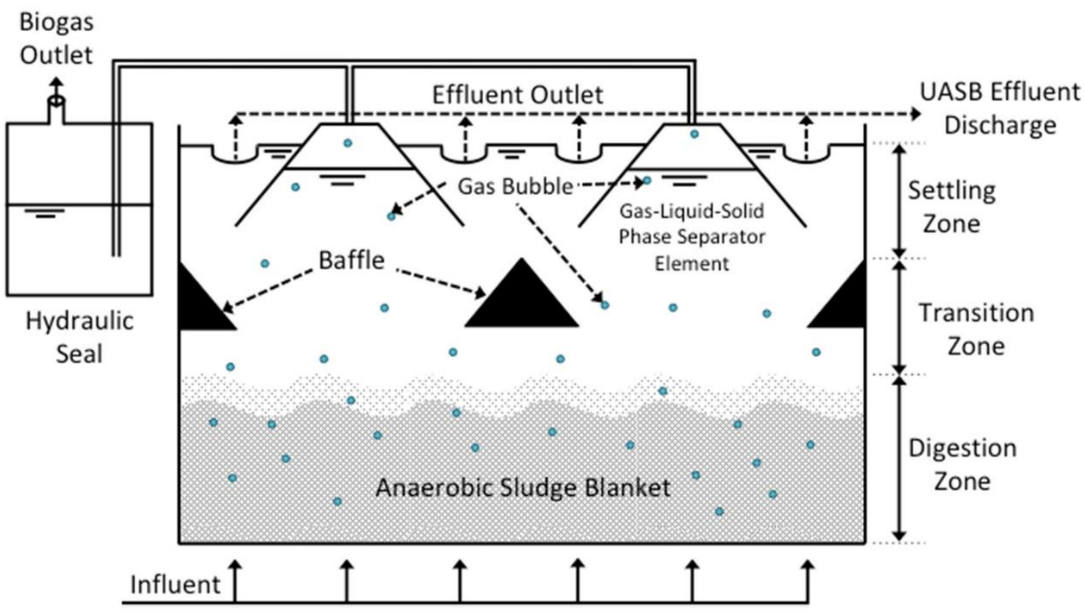

Figure 19 Upflow Anaerobic Sludge Blanket (UASB) reactor

provides a variety of upsides over aerobic treatment, including reduced energy input of the system and decreased creation of surplus sludge per unit mass of stabilised organic matter. Biogas (a useful source of energy) is produced when waste organic materials degrade and decreased biological production reduces the need for nutrients. Anaerobic digestion, on the other hand, is a slow pace process that is reliant on the temperature and pH of waste water. The UASB process is also hampered by toxic contaminants.

Because of the UASB process's cheap capital and operating & maintenance costs as well as low energy requirements, India places a specific emphasis on it. The Upflow Anaerobic Sludge Blanket (UASB) reactor maintains a high concentration of biomass through formation of highly settleable microbial aggregates. The wastewater flows upwards through a layer of sludge. At the top of the sludge blanket layer phase separation between gas-solid-liquid takes place. Any biomass leaving the reaction zone is directly recalculated from the settling zone. The process is suitable for both soluble wastes and those containing particulate matter.

2.7.4 TERTIARY TREATMENT

Tertiary treatment is the final treatment meant for abolishing the secondary effluents and removal of fine suspended solids, traces of organics and bacteria. The sewage effluent from secondary treatment plant is introduced into a flocculation tank where lime is added to eliminate calcium phosphate.

The solution then enters the NH_3 stripping tower. Nitrogen present in waste water exists as NH_4^+ which is converted to gaseous ammonium ion at high pH (11). Phosphorus is removed by adding ferric chloride or aluminium sulphate. The remaining organic materials are removed by desalination, ion exchange and finally chlorination is used for disinfection. The toxic, non-biodegradable chemicals in industrial waste water can be removed by adsorption (on activated charcoal), ion exchange, ultra-filtration, reverse osmosis and electrodialysis.

CHAPTER

3. SCHEME AND ITS COMPONENTS

3.1 UASB TECHNOLOGY

We are adopting UASB based Technology Sewerage Treatment Plant (STP) for treatment of domestic waste water due to its advantages as discussed earlier. This technology was adopted for the first time in India at Jajmau, Kanpur for the 5 MLD STP under Ganga Action Plan Phase-I in the year 1988-89. The efficiency of this STP was about 65% to 70%. Since the desired BOD and TSS could not be achieved by UASB process alone, so it was recommended to provide some additional treatment units, like polishing pond, aerated lagoons etc. to achieve desired values of BOD, TSS and MPN for disposal into water body. At present, this technology followed by additional post treatment units by providing pre-aeration tank for sulphide removal and polishing pond to obtain the BOD_5 of the effluent less than 30 mg/L and suspended solids less than 50 mg/L has been proposed for the 345 MLD Sewage Treatment Plant at Bharwara, Lucknow under Gomati Action Plan Phase-II. The capacity of this STP is biggest in the world at present.

It is somewhat modified version of the contact process, based on an upward movement of the liquid waste through a dense blanket of anaerobic sludge. No inert medium is provided in these systems. The biomass growth takes place on the fine sludge particles, which then develop as sludge granules of high specific gravity. The reactor can be divided in three parts (Figure), sludge bed, sludge blanket and three phase separators (gas-liquid-solid, GLS separator) provided at the top of the reactor. The sludge bed consists of high concentration of active anaerobic bacteria (40 – 100 g/L) and it occupies about 40 to 60% of reactor volume. Majority of organic matter degradation (> 95%) takes place in this zone. The sludge consists of biologically formed granules or thick flocculent sludge. Treatment occurs as the wastewater comes in contact with the granules and/or thick flocculent sludge. The gases produced causes internal mixing in the reactor. Some of the gas produced within the sludge bed gets attached to the biological granules. The free gas and the particles with the attached gas rise to the top of the reactor. On the top of sludge bed and below GLS separator, thin concentration of sludge is maintained, which is called as sludge blanket. This zone occupies 15 to 25% of reactor volume. Maintaining sludge blanket zone is important to dilute and further treat the wastewater stream that has bypassed the sludge bed portion following the rising biogas. The GLS separator occupies about 20 to 30% of the reactor volume. The particles that raise to the liquid surface strike the bottom of the degassing baffles, which causes the attached gas bubbles to be released. The degassed granules typically drop back to the surface of the sludge bed. The free gas and gas released from the granules is captured in the gas collection domes located at the top of the reactor.

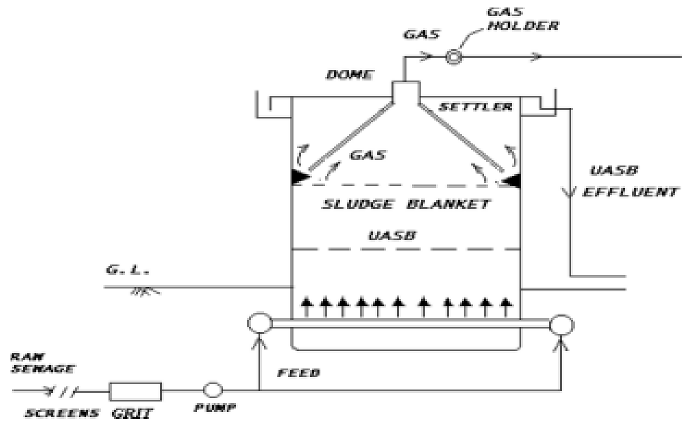

Figure 20 Flow chart of Upflow Anaerobic Sludge Blanket reactor

Liquid containing some residual solids and biological granules passes into a settling chamber, where the residual solids are separated from the liquid. The separated solids fall back through the baffle system to the top of the sludge blanket. The granular biomass from the existing UASB reactor can be used as inoculum material to start-up new UASB reactor. When such material is not available, non-granular material such as anaerobic digested sludge, waste activated sludge and cow dung manure can be used as inoculum. Granular sludge can be developed using non-granular material for inoculation.

Although, there are reports of wastewaters containing high-suspended solids being successfully treated in UASB reactors without primary sedimentation, the separation of suspended solids is still suggested, especially for reactors having non-granular configuration. Pre-treatment such as sedimentation, neutralization of wastewater is normally desirable in treating waste in UASB reactor. Organic loading in the range of 1-20 kg COD/m^3/d can be applied with removal efficiency of 75 to 85 % and HRT of 4 to 24 h.

3.1.1 APPLICATION OF UASB REACTOR FOR WASTEWATER TREATMENT

3.1.1.1 Modes of Operation

UASB reactor is also applicable for the treatment of wastewater from the industries, which are seasonal in origin, like food processing industries. Once, the primary start-up of the reactor is over, with development of good quality of granular sludge, the shutdown of the reactor is possible when the season is over. The reactor put into operation in new season takes very less time (1 to 2 weeks) for this secondary start-up, to restore its COD removal efficiency. For short duration of shut down less than a month reactor can capture its original COD removal efficiency within a week.

3.1.1.2 Treatment Flow Sheet

The typical flow chart required for UASB type wastewater treatment plant is as follows:

1. Pumping,
2. Screening,
3. Grit removal,
4. Skimming Tank (Optional),
5. UASB reactor,
6. Gas collection system,
7. Post-treatment such as aerobic processes or settling tank, and
8. Sludge drying beds.

The provision of screens and grit chamber is necessary for the treatment of municipal wastewater as required in conventional wastewater treatment plants. For certain industrial wastewaters provision of screens and grit chamber may not be necessary. When the wastewater contains floating matter such as, oil, grease, soap, pieces of cork and wood, vegetable debris and fruit skins, it is advantageous to have a skimming tank to remove these materials. The presence of oil and grease, if gets adsorbed on the sludge surface, can hinder transport of metabolites and mass transfer, ultimately causing reduction in process efficiency. This may be accomplished in a separate tank or can be combined with primary sedimentation when wastewater also has high suspended solids of inorganic origin.

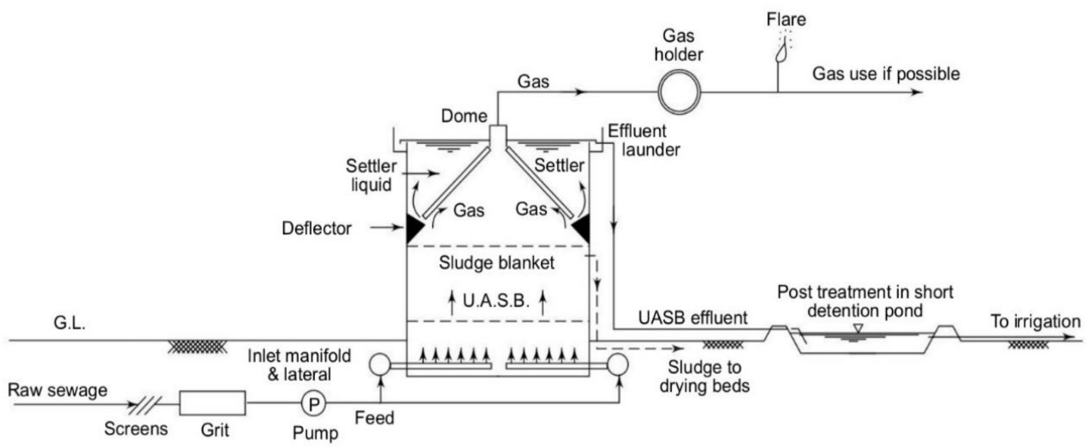

Figure 21 Flow chart of treatment scheme showing screening and grit removal followed by UASB with pumped feed from bottom and further treatment in a short detention pond prior to irrigation.

After the primary treatment, it is required to provide pumping unit to pump the wastewater in upward direction in UASB reactor. Location at which topography of the site suits for utilization of gravity head, choosing appropriate site for UASB reactor may not require pumping. The separate gas collection system can be provided if the gas produced is desired to use for combustion or power generation. Generally, the production of gas in UASB reactor is in the range of 0.25 to 0.35 m^3 CH_4/kg COD removed. The utilization of biogas for power generation is economical for larger treatment plants.

3.1.1.3 Post Treatment

The UASB reactor is an efficient process for removal of organic material and suspended solids from sewage or industrial effluents. Particularly, this process is more attractive for treatment of sewage in warm climate. However, the UASB reactor can hardly remove macronutrients (nitrogen and phosphorous), and pathogenic microorganisms are only partially removed. Hence, depending on the final disposal of the effluent quality, post-treatment may be required for removal of suspended solids, organic matter, nutrients, and pathogens present in the raw wastewaters.

After UASB reactor, some form of post treatment is generally desirable depending upon source of effluent discharge. UASB reactor can hardly remove any nitrogen from the wastewater. Hence, effluent from UASB reactor is suitable for irrigation purposes. UASB reactor when followed by post treatment such as aeration and/or sedimentation could conveniently achieve irrigation standards. The aeration can be obtained to the effluent flowing through a channel to an irrigation area. Where, the treatment efficiency is adequate to meet the discharge standards, further treatment such as, aeration is only necessary to destroy anaerobicity. In such cases simple cascade type aerator can serve the purpose. In some cases where treatment efficiency is meeting the discharge standards for organic matter but the effluent is high in suspended solids, the use of secondary settling tank becomes essential.

When stricter effluent standards have to be met (as for river discharge) some better form of post-treatment may become necessary. The use of aerobic biological treatment is generally preferred for this polishing treatment. The aerobic process such as, bio towers, conventional activated sludge process, or extended aeration can be employed as a second stage treatment. Where the effluent from UASB reactor is expected to be high in nutrient such as nitrogen and phosphorous, the post treatment needs to be designed for removal of these nutrients to meet discharge standards for surface water. Shallow oxidation ponds can also be used after UASB reactor for complete treatment of wastewater.

3.1.1.4 Design Procedure for UASB Reactor

The UASB reactor can be designed as circular or rectangular. Modular design can be preferred when the volume of reactor exceeds about 400 m^3. It is necessary to select proper range of operating parameters for design, such as OLR, SLR, superficial liquid upflow velocity (referred as liquid upflow velocity), and HRT. The literature recommendations for all these parameters and design procedure to account these recommendations are given below.

3.1.1.4.1 Organic Concentration and Loading

For COD concentration in the range 2 to 5 g/L, the performance of the reactor depends upon the loading rate and is independent of influent substrate concentration. For COD concentration greater than 5 g/L, it is recommended to dilute the wastewater to about 2 g COD/ L during primary start-up of the reactor. Once, the primary start-up of the reactor is over with granulation of sludge, loading rates can be increased in steps to bring the actual COD concentration of the wastewater. The loading above 1 - 2 kg COD/ m^3/d is essential for proper functioning of the reactor. For sewage treatment, the design of reactor at higher loading rate is not possible due to limitations of upflow velocity, and maximum loading of about 2 to 3 kg COD/m^3/d can be adopted for design.

Table 7 Recommended loading range for design of UASB reactor based on COD concentration at average flow

Category of wastewater	COD concentration, mg/L	OLR, Kg COD/ m^3.d	SLR, Kg COD/ kg VSS.d	HRT, Hours	Liquid Upflow Velocity, m/h	Expected Efficiency, %
Low strength	Up to 750	1.0 - 3.0	0.1 - 0.3	6 – 18	0.25 – 0.7	70 –75
Medium strength	750 – 3000	2.0 – 5.0	0.2 – 0.5	6 – 24	0.25 – 0.7	80 – 90
High strength	3000 – 10,000	5.0 – 10.0	0.2 – 0.6	6 – 24	0.15 – 0.7	75 – 85
Very high strength	> 10,000	5.0 – 15.0	0.2 – 1.0	> 24	---	65 – 75

(Source: Ghangrekar et al., 2003)

3.1.1.4.2 Reactor Volume

Based on the higher suitable value of OLR, for given COD concentration, the volume of reactor required is to be worked out as:

Volume = (Flow Rate * COD concentration) / OLR

If the volume is not meeting the requirements, the OLR can be reduced to increase the volume. The volume of the reactor is thus, finalized to meet both the requirements. For this volume, the HRT should not be allowed to be less than 6 h for any type of wastewater and generally, it should be less than 18 h to reduce volume and hence, cost of the reactor. For very high strength of the wastewater, COD greater than 10,000 mg/L, it may not be possible to meet this requirement, hence, under such situation the HRT may be allowed to exceed even 24 h and as high as 200 h.

3.1.1.4.3 Superficial Liquid Upflow Velocity

Higher upflow velocities, favours better selective process for the sludge and improve mixing in the reactor. However, at very high upflow velocity, greater than 1.0 to 1.5 m/h, the inoculum may get washed out during start-up or during normal operation granules may get disintegrated, and the resulting fragments can easily wash out of the reactor. The maximum liquid upflow velocity allowed in design should not exceed 1.2 – 1.5 m/h. Upflow velocities as 0.25 to 0.8 m/h are favourable for granule growth and accumulation, during normal operation of the reactor and maximum upflow velocity up to 1.5 m/h at peak flow conditions for short duration can be used in design.

3.1.1.4.4 Reactor Height and Area

The reactor should be as tall as possible to reduce plan area and to reduce cost of land, GLS device, and influent distribution arrangement. The height should be sufficient to provide enough sludge bed height to avoid channelling and to keep liquid upflow velocity within maximum permissible limits. In order to minimise channelling the minimum height of the sludge bed should be about 1.5 to 2.5 m. For this reason, the minimum height of the reactor should be restricted to 4.0 m, to conveniently accommodate sludge bed, sludge blanket and GLS separator. The maximum height of the reactor can be about 8 m. The height of the reactor adopted in practice is usually between 4.5-8 m and 5 m is the typical height used for UASB reactors.

While designing, initially suitable height of the reactor (about 5m) can be chosen, and superficial liquid upflow velocity is to be worked out as height/HRT. It is recommended to adopt upflow velocity of 0.5 m/h at average flow and 1.0 m/h to 1.2 m/h at peak flow. Accordingly, if the upflow velocity exceeds the maximum limits height of the reactor can be reduced in steps up to minimum of 4 to 4.5 m. If this is not possible in the applicable range of height, HRT shall be modified and fresh reactor volume and OLR shall be worked out. For low strength wastewater, the maximum liquid upflow velocity becomes limiting and for very high strength wastewater very low velocity (less than 0.1 m/h) is required while designing the UASB reactor. Under certain situations, the revised OLR may be less than the initial OLR recommended. It is advisable to allow lowering of OLR in such situations to control upflow velocity in the reactor for proper performance of the reactor.

3.1.1.4.5 Gas-Liquid-Solid (GLS) Separator

In order to achieve highest possible sludge hold-up under operational conditions, it is necessary to equip the UASB reactor with a GLS separator device. The main objective of this design is to facilitate the sludge return without help of any external energy and control device. The GLS should be designed to meet the requirements such as, provision of enough gas-water interface inside the gas dome, sufficient settling area outside the dome to control surface overflow rate; and sufficient aperture opening at bottom to avoid turbulence due to high inlet velocity of liquid in the settler, and to allow proper return of solid back to the reactor. Due attention has to be paid to the geometry of the unit and its hydraulics, to ensure proper working of the GLS separator.

Aperture width at bottom of gas dome: The area of aperture (A_p) required can be computed based on the maximum inlet velocity of liquid to be allowed. This area can be estimated as flow rate per dome for rectangular reactor (or central dome in case of circular) divided by maximum velocity to be allowed. The maximum inlet velocity of 3 m/h is safe for medium and high strength wastewater and for low strength wastewater lower inlet velocity should be preferred. The width of aperture (W_a) is to be calculated as aperture area divided by length (or in case of circular reactor by diameter) of the reactor. It is recommended to use minimum aperture width of 0.2 m and if the width required is greater than 0.5 m, then increase the number of domes by one and repeat earlier steps till it is less than 0.5 m.

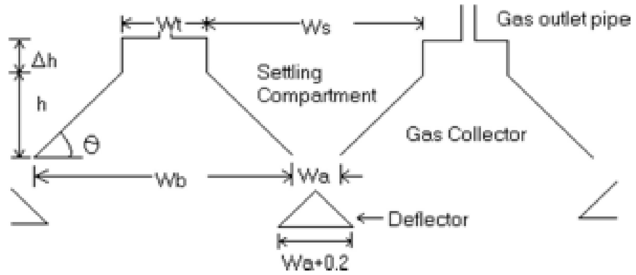

Figure 22 Details of the Gas-Liquid-Solid (GLS) Separator

Check for Surface overflow rate: The width of the water surface (W_s) available for settling of solids for each gas dome, at top of the reactor, can be calculated as difference of base width of dome ($WB = W_b + W_a$) and W_t. The corresponding surface overflow rate is calculated as hydraulic flow rate per dome divided by product of length (or diameter) and W_s. It is recommended that the surface overflow rate for effective settling of solids back to the reactor should be less than 20 $m^3/m^2/d$ at average flow and should be less than 36 $m^3/m^2/d$ under peak flow conditions. If the calculated surface overflow rate is meeting these criteria the design of the GLS separator is final. When it is exceeding the limits recommended, it is advisable to reduce the height of the reactor, thus, for same volume of the reactor more plan area will be available. When the height of the reactor is reduced all earlier steps for design of GLS separator should be repeated to satisfy all design criterions. The minimum height of the reactor should be restricted to 4.0 m (preferably 4.5 m). Once, all the design criteria are satisfied the angle of inclination of the gas collector dome with horizontal (angle) can be calculated as = $\tan^{-1}[\, 2h/ (W_b - W_t)]$.

Baffle of sufficient overlap (0.1 to 0.2 m) should be provided below the gas collector in order to avoid entry of biogas in the settling compartment. The diameter of the gas exhaust pipes should be sufficient to guarantee easy removal of the biogas from the gas collection cap, particularly in case of foaming. Generally, lower reactor height is required for UASB reactor treating sewage. Under certain situation, particularly for very low strength of wastewater, even with reduction of height to the minimum may not meet all design requirements. In such cases the OLR adopted for design can be reduced to provide greater volume of the reactor and hence more plan area to meet the entire design criterion.

3.1.1.4.6 Effluent Collection System

The effluent has to leave the UASB reactor via number of launders distributed over entire area discharging to main launder provided at periphery of the reactor. The effluent launders can be designed in such a way that the weir loading ($m^3/m/d$) should not exceed the design criteria of secondary settling tank. The width of the launders may be minimum 0.20 m to facilitate maintenance. The depth of the launder can be worked out as open channel flow. Additional depth of 0.10 to 0.15 m shall be provided to facilitate free flow. On both sides of the launders 'V' notches shall be used. When effluent launders are provided with scum baffles, the 'V' notches will be protected from clogging as the baffles retain the floating materials.

3.1.1.5 Design of Feed Inlet System It is important to establish optimum contact between the sludge available inside the reactor and wastewater admitted, and to avoid channelling of the wastewater through sludge bed. Hence, proper design of inlet distribution system is necessary. Depending on topography, pumping arrangement, and likelihood blocking of inlet pipes, one could provide either (i) gravity feed from top (preferred for wastewater with high suspended fraction), or (ii) pumped feed from bottom through manifold and laterals (preferred in case of soluble industrial wastewaters). In general, the area to be served by each feed inlet point should be between 1 and 3 m^2.

3.1.1.5.1 Other Requirements

It is necessary to keep provision for removal of excess sludge from the reactor. Although, the excess sludge is wasted from about middle height of the reactor, it is also necessary to make arrangement at bottom of the reactor. In addition, 5 to 6 numbers of valves should be provided over reactor height to facilitate sampling of the sludge. For treating high strength wastewater, it is recommended to apply effluent recycle, in order to dilute COD concentration and to improve contact between sludge and wastewater. For treating wastewater with COD concentration greater than 4 - 5 g/L, it is recommended to apply dilution during start-up, for proper granulation of sludge inside UASB reactor. Auxiliary equipment has to be installed for addition of essential nutrients, and alkalinity for control of pH of the influent. The other equipments to be provided are for measurement of pH, temperature, influent flow rate, and gas production rate.

CHAPTER

4. METHODOLOGY & DESIGN

Methodology for the project can be divided into two parts: -

1. Characterization of waste water parameters
2. Design of UASB based STP units.

4.1 CHARACTERIZATION OF WASTE WATER PARAMETERS

Methods of sampling and test (Physical and Chemical) of Waste Water: - (IS 3025)

4.1.1 Methodology for Measurement of pH Value (IS 3025: Part 11)

The pH of a sample is nothing but the negative logarithm of concentration of hydrogen ion. pH fluctuates from 6-8 in sample of waste water, due to hydrolysis of salts of acids and bases. Hydrogen Sulphide, Carbon dioxide and Ammonia which are dissolved upset pH value of water. pH value may be greater than 9 in alkaline springs and the pH equal or less than 4 for acidic ones. Industrial and manufacturing wastes do upset the pH as it depends on buffer capacity of water. pH value of water sample in lab changes because of loss or, reactions with sediments, absorption of gases, chemical reaction taking place within the sample bottle. Therefore, pH value should sooner be evaluated at the time of collection of samples. pH can be determined electrometrically or calorimetrically.

Principle

The pH of the waste water sample is determined electrometrically using either a glass electrode in combination with a reference potential or a combination electrode.

Apparatus used

- pH meter – With glass and reference electrode
- Thermometer

Procedure

The instrument with a buffer solution of pH near that of the sample is standardized and electrode against at least one additional buffer of different pH value is checked. The temperature of the water is measured and if temperature compensation is available in the instruments, it is adjusted accordingly. After the standardization place the sample in the beaker and immerse the electrode, then take the reading in the pH meter and the temperature.

4.1.2 Methodology for Measurement of Turbidity (IS 3025: Part 10)

Principle

It is centred on comparison of the intensity of light scattered by the sample under convinced conditions with the intensity of light scattered by a standard reference suspension under the similar conditions.

Apparatus used

- Sample tubes –clear and colourless sample tubes are taken
- Turbidity meter: Systronics digital nephlo-turbidity meter.

Procedure

- Three pin plugs was inserted into appropriate 230V AC mains socket.
- The instrument was switched on and was allowed 10-15 minutes to warm up.
- An appropriate range was selected.
- A CALIB control was set to the maximum in clockwise position.
- The test tube was inserted in distilled water into cell holder and was covered with light shield.
- A SET ZERO control was adjusted to get zero on the display.
- The test tube was adjusted and the test tube containing standard solution was replaced. And CALIB control was set up for display the result.
- The instrument now ready for test samples test tube containing unknown sample in cell holder is inserted. The display directly gives the turbidity in NTU.

4.1.3 Methodology for Measurement of Biochemical Oxygen Demand (IS 3025: Part-44)

Principle

Bio-chemical Oxygen Demand (BOD) is a measure of the oxygen used by microorganisms to decompose organic waste. If there is a bulky quantity of organic waste in the water supply, a lot of bacteria will also be present to decompose this waste. In this case oxygen demand will be high (due to all the bacteria) so the BOD level will be high. As the waste is consumed or detached through the water, BOD levels will begin to decline.

Apparatus used

- Incubation bottles- 300 mL capacity
- Magnetic stirrer

Procedure

The unknown sample was placed in the incubation bottle where 4 capsules (4 gm.) of NaOH had been kept at the neck of the bottle. A magnetic stirrer was continuously rotated inside the bottle. Then it is kept air tight by means of special caps attached with an electronic meter. The electronic meter directly gives records BOD reading at every 24 hours. Now the bottles are preserved in the Refrigerator for days as per requirement of study. The same procedure can be adopted for BOD 3 days and BOD 5 days).

4.1.4 Methodology for Measurement of Dissolved Oxygen (IS 3025: Part 38)

Principle

The Winkler Method uses titration to determine dissolved oxygen in the water sample. A bottle is filled completely with water such that no air is left to skew the results. The dissolved oxygen in the sample was then "fixed" by adding a series of reagents which forms an acid compound which is then titrated by means of a neutralizing compound that causes the colour change. The colour change point is called the "endpoint," which coincides with the dissolved oxygen concentration.

Reagent list

- 2mL Manganese sulphate
- 2mL alkali-iodide-azide
- 2mL concentrated sulfuric acid
- 2mL starch solution
- Sodium thiosulfate

Procedure

- Carefully a 300-mL glass Biological Oxygen Demand (BOD) stoppered bottle was filled brim-full with sample water.
- Immediately 2mL of manganese sulphate was added to the collection bottle by inserting a calibrated pipette just beneath the surface of the liquid. (If the reagent is added above the sample surface, oxygen will be introduced into the sample.) The pipette was slowly squeezed so that so no bubbles are introduced via the pipette.
- 2 mL of alkali-iodide-azide was added reagent in the same manner.
- The bottle was closed with stopper with care to be sure no air is introduced. The sample was mixed by inverting several times. Air bubbles were checked; the sample was discarded and started over if any are seen. If oxygen is present, a brownish-orange cloud of precipitate or floc would appear. When this floc would have settled to the bottom, the sample was mixed by turning it upside down several times and let it settle again.
- 2 mL of concentrated sulfuric acid was added through a pipette. Carefully stopper was placed and was inverted several times to dissolve the floc. At this point, the sample has become "fixed" and can be stored in a cool dark place for up to 8 hours. As an added precaution distilled water was squirted along the stopper as a

precaution, and the bottle was capped with aluminium foil and also with a rubber band during the storage period.

- 200 mL of the sample was titrated with sodium thiosulfate to obtain a pale straw colour.
- Titration was done by slowly releasing the titrant solution from a standard calibrated pipette into the flask and continuously swirling or stirring the sample water.
- 2 mL of starch solution was added to get a blue colour form.
- Titration was continued until the sample blue colour decolorizes. Care has to be taken to ensure that each drop is carefully mixed with sample before the next drop is added. The number of millilitres of the titrant used gives directly the amount of dissolved oxygen in mg/L.

4.1.5 Methodology for Measurement of Hardness (IS 3025: Part 21)

Principle

Total hardness measures the amount of calcium and magnesium and is expressed in terms of calcium carbonate. As, to remain healthy, our body needs both Calcium and Magnesium. If water would be too hard then it would not only decrease the washing capability of many soaps and detergents but will also affect the taste of the water.

Apparatus used

- A pipette – (Minimum 50 mL capacity)
- Glass bottle (4-8 oz.)

Procedure

- 50 mL of the sample was pippeted into 4 – 8 oz. glass bottle.
- The standard soap solution was added in small portions at first (0.5 mL), shaking vigorously after each addition.
- As the end point comes closer, the quantity added should be decreased by 0.1 mL for each addition.
- A perpetual lather is created which is last for 5 minutes with the bottle on its side, and the mL of soap solution used was recorded.
- The soap solution was added in small quantities. If the lather again vanishes, the first point obtained was incorrect due to the existence of magnesium salts. (The volume of soap used to obtain this false end point may be used for calculation of the approximate magnesium hardness by substituting in the formula used for calculation).

- The soap solution was continuously added until the true end point is reached and volume used was recorded.
- Sample is diluted to 50 mL with freshly boiled and cooled distilled water.

4.1.6 Methodology for Measurement of Alkalinity (IS 3025: Part 23)

Apparatus used

1) Pipette – Minimum 100 mL capacity
2) Erlenmeyer flask
3) A burette

Reagents used

1) Phenolphthalein indicator
2) 0.02N sulfuric acid
3) Methyl orange indicator

Procedure

1. 100 mL of the sample was pipetted into the Erlenmeyer flask and the same quantity of distilled water was taken into another.
2. 3 drops of phenolphthalein indicator were added to each flask.
3. If the sample shows pinkish colour 0.02N sulfuric acid was added from a burette until the pink colour just vanishes and no. of mL of acid used was recorded.
4. 3 drops of methyl orange indicator were added to each flask.
5. If the sample becomes yellow in colour, 0.02N sulfuric acid is added until the first difference in colour is noted in comparison with the distilled water. If the end point is orange the no. of mL of acid used is recorded.

4.1.7 Methodology for Measurement of Acidity (IS 3025: Part 22)

Apparatus used

1) Pipette – Minimum 100 mL capacity
2) Erlenmeyer flask
3) A burette

Reagents used

1) Phenolphthalein indicator
2) 0.02N sodium hydroxide

Procedure

1) 100 mL of the sample was pippeted into an Erlenmeyer flask.
2) Then 3 drops of phenolphthalein indicator were added.

3) 0.02N sodium hydroxide from burette was added until the first permanent pink colour appears and the no. of mL of sodium hydroxide used was recorded.

Calculation

mL of 0.02N sodium hydroxide × 10 = p.p.m. total acidity expressed in terms of $CaCO_3$

4.1.8 Methodology for Measurement of Total Solids (IS 3025: Part 15)

Apparatus used

1) Pit crucible
2) A desiccator
3) Flask or pipette

Procedure

1) The pit crucible was cleaned and was placed in a 103°Celsius oven for 1 hr.
2) The crucible was placed in a desiccator until cools, and then it was weighed.
3) The sample thoroughly was mixed and was measured as 100 mL by volumetric flask or pipette.
4) The sample was transferred to the dish and the flask was rinsed or pipetted several times with small portions of distilled water and the rinsing was added to the dish. It has to be making sure that all suspended matter is completely transferred to the crucible.
5) After the sample is evaporated, the crucible is dried and residue in the 103°C oven for 1 hr., cool in the desiccator and weigh.

Calculation

[Increase in weight (cm) × 1000] ÷ ML of sample = ppm total solids

4.1.9 Methodology for Measurement of Chloride Present (IS 3025: Part 32)

Apparatus used

1) Porcelain evaporating dish
2) A burette
3) A pipette

Reagents used

1) Potassium chromate indicator
2) Standard silver nitrate solution

Procedure

1. 50 mL of the sample was pippeted in the porcelain evaporating dish.
2. The same quantity of distilled water was placed into second dish for colour comparison.

3. 1 mL of potassium chromate indicator was added to each.
4. Standard silver nitrate solution was added to the sample from a burette, a few drops at a time, with constant alternating until the first permanent reddish coloration appears. This can be determined by comparison with the distilled water. The mL of the silver nitrate solution used was recorded.
5. If more than 7 or 8 mL of silver nitrate solution is required, the entire procedure should be repeated using a smaller sample diluted to 50 mL with distilled water.

Calculation

[(mL of silver nitrate used – 0.2) × 500] ÷ mL of sample = p.p.m. chloride

DESIGN

4.2 DESIGN OF UASB BASED STP UNITS

4.2.1 DESIGN PARAMETERS

Design average flow = 10 million litre per day (MLD)

Peak factor = 2.0

4.2.2 DESIGN OF PUMP HOUSE

Average Discharge,

$$Q_{avg} = 10 \text{ MLD} = 10^4 \text{ m}^3/\text{d} = 0.116 \text{ m}^3/\text{s}$$

Max. sewage flow,

$$Q_{max} = 2 * 0.116 = 0.232 \text{ m}^3/\text{s}$$

Assuming Min. flow as 40% of average flow,

$$Q_{min} = 0.4 * 0.116 = 0.046 \text{ m}^3/\text{s}$$

▶ Design of rising main:

Assuming 1 m/s as the flow through velocity in the rising main, the cross-sectional area of the rising main at peak flow, A = flow/velocity

$$A = 0.232 \text{ m}^2$$

Diameter of the rising main, $d_r = \sqrt{(4 * A)/\pi} = 0.54$ m

Provide a rising main of 550 mm diameter and area = 0.23 m^2

▶ Design of sump well (wet well):

Assuming that the sump well retains the flow of waste water for 20 min., the maximum quantity of sewage in the sump,

$$Q_{max} * t = 0.232 * 20 * 60 = 278.4\, m^3$$

Quantity of sewage in a rising main of 200 m length $= 200 * 0.23 = 46\, m^3$

Total Capacity of the sump required $= 278.4 + 46 = 324.4\, m^3 = 325\, m^3$

Assuming a max. of 4 m depth of the sump, the surface area of the sump,

$$A_s = \frac{325}{4} = 81.25 \text{ m}^2$$

Providing 3 sump wells for 3 pumps, 2 working and 1 standby (2W+1S), such that 2 wells will always remain in operation when one of the wells is taken under repair or maintenance.

The surface area of one sump is $\frac{81.25}{2} = 40.625\, m^2$

Diameter of each sump, $d_s = \sqrt{4 * 40.625/\pi} = 7.2\, m$

To determine minimum liquid depth in the sump well, Q_{min} in the well + flow in rising main

Assuming 20 min. detention time,

$$Q_{min} * t = 0.046 * 20 * 60 = 55.2\ m^3$$

Minimum volume of liquid in the well = $55.2 + 46 = 101.2\ m^3$

Area of one wet well $= \frac{\pi}{4} * 7.22 = 40.7\ m^2$

Minimum liquid depth in each well $= \frac{101.2}{40.7} = 2.5\ m$

Design of Dry well: A dry well is designed on the basis of the space equal to that is required for housing the working as well as standby pumps. Normally one can assume the same dimensions as that for a wet well. Additional space is provided for ladder and working area of maintenance.

▶ <u>Design of pumps</u>:

Providing 3 pumps, 2 in operation and 1 as standby, the maximum flow that each pump has to lift, $Q_{pump} = \frac{0.232}{2} = 0.116\ m^3/s$

Frictional losses in pipe,

$$h_f = \frac{flv^2}{2gd} = \frac{0.01*200*12}{2*9.81*0.54} = 0.19\ m \qquad \text{(Assuming } f = 0.01\text{)}$$

Assuming losses in bends, etc. = 0.5 m

Total Head, $H = Pumping\ Head + Head\ losses$

Assuming sewage is to pumped against a head of 8 m,

$$H = 8 + 0.19 + 0.5 = 8.69\ m \approx 9\ m$$

Power required $= \frac{W*Q*H}{75*\eta_p*\eta_m} = \frac{1000*0.116*9}{75*0.7*0.8} = 24.8$ BHP $\approx 25\ BHP$

(Assuming $\eta_p = 70\%,\ \eta_m = 80\%$)

Total Power required $= 2 * 25 = 50\ BHP$

Provide 3(2W+1S) pumps, each of 25 BHP for each well.

▶ <u>Design of Suction Pipe</u>:

Max. flow in the pipe for each wet well $= 0.116\ m^3/s$

Assuming velocity of flow in the pipe, $v = 1\ m/s$, the cross-sectional area of the intake pipe, $A_x = \frac{0.116}{1} = 0.116\ m^2$

Diameter of suction or inlet pipe, $d_{in} = \sqrt{4 * 0.116/\pi} = 0.38\ m \approx 0.4\ m$

Provide 400 mm diameter pipes.

▶ Approximate length of pump house $= 7.2 + 7.2 + 7.2 \approx 25\ m$

▶ Approximate width of pump house $= 7.2 + 7.2 + 5 \approx 20\ m$

4.2.3 DESIGN OF APPROACH CHANNEL

Providing two channels in one unit

$$Q_{\max} = 20/2 = 10\, MLD = 10^4\, m^3/d = 0.116\, m^3/d$$

$$A_x = \frac{0.116}{1} = 0.116\, m^2 \qquad \text{(Assuming } v = 1\, m/s\text{)}$$

$$b/d = 1.5/1 \qquad \Longrightarrow 1.5 * d^2 = 0.116$$

$$d = 0.278\, m \approx 0.3\, m$$

$$b = 1.5 * 0.3 = 0.45\, m$$

Provide net depth, d $= 0.3\, m$ and width, b $= 0.45\, m$

Area $= 0.3 * 0.45 = 0.135\, m^2$

Assuming free board is 0.3 m, the total depth of tank, $D_f = 0.3 + 0.3 = 0.6\, m$

Assuming length of the channel as 2 m, so provide a 2 m long channel of 0.45 m width and 0.6 m depth

$$\text{Dimension}-(2 * 0.45 * 0.3 + 0.3\, FB)$$

Assuming slope of channel as 1:1000 and $n = 0.012$ for cement concrete (Manning's roughness coefficient)

Wetted perimeter of the channel $= 2D + B = 2 * 0.3 + 0.45 = 1.05\, m$

Hydraulic mean radius, $R = \frac{A}{P} = \frac{0.135}{1.05} = 0.1286\, m$

$$v_h = \frac{1}{n} * R^{2/3} * s^{1/2} = \frac{1}{0.012} * (0.1286)^{2/3} * (0.001)^{1/2} =$$

$$0.671\, m/s < v_h = 1\, m/s \qquad \text{(Assumed)}$$

Again, Assuming slope of channel as 1:250 and n = 0.015

$$v_h = \frac{1}{0.015} * (0.1286)^{2/3} * (1/250)^{1/2} = 1.074\ m/s > v_h = 1\, m/s \qquad \text{(Assumed)}$$

Hence OK

Check for flow at one-third depth $D/3 = 0.3/3 = 0.1\, m$

Cross sectional area $= 0.1 * 0.45 = 0.045\, m^2$

Wetted perimeter = $2D + B = 2 * 0.1 + 0.45 = 0.65\ m$

Hydraulic mean radius $= \frac{A}{P} = \frac{0.045}{0.65} = 0.0692\ m$

Velocity at $\frac{D}{3} = \left(\frac{1}{n}\right) * R^{2/3} * S^{1/2} = 1/0.015 * (0.0692)^{2/3} * (1/250)^{1/2}$

$(v_h)_{d/3} = 0.711\ m/s > 0.5\ m/s$ (min. velocity)

4.2.4 DESIGN OF SCREEN CHAMBER

$$Q_{max} = 20\ MLD = 0.232\ m^3/s$$

Dimension of screen chamber = dimension of approach channel $= 2\ m * 0.45\ m * 0.3\ m$

Cross sectional area, $A_x = B * D = 0.45 * 0.3 = 0.135\ m^2$

Approach velocity, $v_h = 0.232/(2 * 0.135) = 0.86\ m/s$

▶ <u>Number of bars</u>:

Provide bars of 10 mm × 50 mm with a clear opening of 25 mm. Let n be the number of bars then, Width of opening (n+1) + width of bars (n) = B (total width)

$$0.025 * (n + 1) + 0.01 * n = B = 0.45$$

n =12

Provide 12 bars of 10 mm × 50 mm with 25 mm clear spacing

Total width $= 12 * 0.01 + 13 * 0.025 = 0.445\ m$

Effective width of channel, $B_e = 0.45 - 12 * 0.01 = 0.33\ m$

Effective cross-sectional area, $B_e * D = 0.33 * 0.3 = 0.099\ m^2 \approx 0.1\ m^2$

Velocity of flow through screen bars, $v = \frac{flow}{(effective\ cross\ sectional\ area)}$

$$v = \frac{0.116}{0.1} = 1.16\ m/s \approx 1.2\ m/s$$

Head loss, $h_L = 0.0729 * (v^2 - v_h^2)$

Here, $h_L = 0.0729 * (1.2^2 - 0.86^2) = 0.051\ m$ (OK)

Assuming screening production as $0.0012\ m^3/ML$ of flow for 25 mm opening and a flow of 10 MLD for channel.

Quantity of screening produced $= 0.0012 * 10 * 2 = 0.024\ m^3/d$ (for 2 channel)

Cleaning can be done manually every 3 to 4 days from two channels.

▶ Design of perforated plate:

Provide length of the plate equal to the total width of the chamber with position wall of 0.2 m thick, which is 1.1 m $(0.45 * 2 + 0.2)$.

Assuming width of the plate equal to 0.3 m and depth of pocket equal to 0.2 m for collecting screenings.

Capacity of the screening pocket, $C = 1.1 * 0.3 * 0.2 = 0.066\ m^3$

Which is approximately equal to 3 days cleaning

Inclined length of bars, $\frac{D}{sin45°} = \frac{0.3+0.3(FB)}{sin45°} = 0.85\ m$

Horizontal projected length of bars $= 0.85 * cos45° = 0.6\ m$

Let the length of the outlet zone be the width of the perforated plate $+ 0.2\ m = 0.3\ m + 0.2\ m = 0.5\ m$

Assuming length of the inlet zone as 0.9 m, the total length of the screen channel is $0.9 + 0.6 + 0.5 = 2\ m$

$$\Rightarrow \text{Size of perforated plate} = 0.45\ m * 0.3\ m$$

$$\Rightarrow \text{Screening collection pocket} = 0.45\ m * 0.3\ m * 0.2\ m$$

4.2.5 DESIGN OF GRIT CHAMBER

Assume no. of channels in one unit = 2

Detention time = 30 s

$$Q_{\max} = 20{,}000\ m^3/d$$

Max. flow per channel $= \frac{20000}{2} = 10000\ m^3/d$

Vol. of each channel, $V = \frac{10000\ m^3}{d} * \frac{30\ s}{\frac{24hr}{d} * \frac{60min}{hr} * \frac{60s}{min}} = 3.5\ m^3$

Assuming flow through velocity, $v_h = 0.2\ m/s$

Length of each channel $= 0.2 * 30 = 6\,m$

Cross sectional area of channel, $A_x = \frac{V}{L} = \frac{3.5}{6} = 0.6\,m^2$

Assuming width of channel, B = 1 m, liquid depth in channel, $D = \frac{A_x}{B} = 0.6\,m$

Therefore, net dimensions $L = 6\,m,\ B = 1\,m,\ D = 0.6\,m$

Check for surface loading rate: Surface area of channel, $A_s = 6 * 1 = 6\,m^2$

At Q_{peak}, $SLR = \frac{10000}{6} = 1666.67\,m/d$ (within acceptable limits)

At Q_{avg}, $SLR = \frac{5000}{6} = 833.33\,m/d$

Therefore, the length and width are acceptable.

Adding 10% for each inlet and outlet zone $= 6 * 0.1 * 2 + 6 = 7.2\,m \sim 7.5\,m$

And total width of the chamber with 2 channels including 0.2 m thick one partition wall and two side walls $= 1 + 1 + 0.2 + 0.2 + 0.2 = 2.6\,m$

Additional Depth is required to be provided for accumulation of grit in the chamber. Assuming grit accumulated to be 0.03 m^3/1000 m^3/d of flow and removal of grit is done at interval of 3 days, volume $= \frac{0.03*10000*3}{1000} = 0.9\,m^3$

Therefore, the net depth of grit storage $= \frac{0.9}{6} = 0.15\,m$

However, provide the depth of 0.2 m for grit storage below the actual crest of weir.

Total Depth,

$$D_{total} = net\ depth + grit\ storage\ depth + freeboard(FB) = 0.6 + 0.2 + 0.3$$

$$D_{total} = 1.1\,m$$

And total liquid depth, $d = 0.8\,m$

Providing 2 channels, the overall dimensions of the grit chamber will be:

$$L_{total} = 7.5\text{ m}$$

$$B_{total} = 2.6\,m$$

$$D_{total} = 1.1\,m\ (with\ 0.3\ FB)$$

$$t = 30\,s$$

4.2.6 DESIGN OF UASB REACTOR (with Excel sheet **QR** code)

Assumed temperature range $= 22°\ to\ 28°C$ and the wastewater characteristics are of follows:

$$pH = 7.3 - 7.7 \qquad BOD = 350\,mg/L$$

$$COD = 820\,mg/L \qquad sulphate = 105\,mg/L$$

$$TSS = 395 mg/L \qquad VSS = 270\,mg/L$$

Raw sewage will be pumped to the inlet chamber at the top of the UASB floor of UASB kept at 3m below the GL. Average flow $= 10000\,m^3/d = 0.116\,m^3/s$

- ▶ <u>Sludge Production</u>:

Assume BOD removal efficiency = 80 % (UASB effluent BOD = 70 mg/L)

VSS produced in BOD removal $= 0.1 * 350 * 0.8 = 28\, mg/L$

Non – degradable residue $= VSS * (1 - 0.4) = 270 * 0.6 = 162\, mg/L$

Ash received in inflow $= TSS - VSS = 395 - 270 = 125\, mg/L$

Total sludge produced $= 28 + 162 + 125 = 315\, mg/L$

- ▶ <u>SRT and HRT values</u>:

$$SRT = \frac{total\, sludge\, in\, reactor}{sludge\, withdrawn\, per\, day}$$

Depth of sludge blanket $= 2.2\, m$, Effectiveness coefficient $= 0.85$

Average conc. of sludge in blanket $= 70\, kg/m^3$

Let full depth of reactor (including settler)$= 5m$

$$SRT = 30\, days = \frac{\left(70 \frac{kg}{m^3}\right)\left(\frac{2.2}{5.0}\right)(0.85)(flow\, rate\, m^3/d)\left(HRT \frac{hr}{24}\right)}{(322\, kg/m^3)(flow\, rate\, m^3/d)}$$

$$HRT = \frac{30 * 315}{(70)\left(\frac{2.2}{5}\right)(0.85)\left(\frac{1000}{24}\right)} = 8.66\, hrs\, at\, average\, flow$$

$$= 4.33\, hrs\, at\, peak\, flow \text{ (Acceptable)}$$

- ▶ <u>Upflow velocity</u>:

Upflow velocity at avg. flow$= \frac{5\, m}{8.66\, hrs} = 0.577\, m/hr$

We took $0.5\, m/hr$ (so that we are able to retain even flocculant sludge within UASB)

Upflow velocity at peak flow $= 0.5 * 2 = 1\, m/hr$

- ▶ <u>Reactor dimensions</u>:

Reactor area required $= \frac{total\, flow\, m^3/d}{upflow\, velocity,\, m/hr(24)} = \frac{10000}{0.5*24} = 833\, m^2$

Provide reactor of 35 m length, 24 m width and 5 m side water depth + 0.5 m freeboard.

Check for organic loading: $Volumetric\, organic\, loading = \frac{(COD\, load)}{volume\, of\, reactor}$

$$= \frac{820*10000*1000}{1000000*(24*35*5)} = 1.95\, kg\, COD/m^3/d$$

Organic loading on sludge blanket $= \frac{820*10000}{1000*0.4\left(\frac{VSS}{TSS}\right)\left(70\, \frac{kg}{m^3}\right)*24*35*2.2}$

$$= 0.158\, kg\, COD/kg\, VSS/d$$

(Acceptable)

▶ <u>COD Removed and Methane Produced</u>:

i. Total COD removed = 80% of incoming load

$$= 0.8 * (\frac{820}{1000}) * 10000 = 6560\ kg/day$$

ii. COD consumed in sulphate reduction:

$$\text{Sulphate incoming} = 105\ mg/L$$

Assuming sulphate removal to be $80\% = 0.8 * 105 = 84\ mg/L$

$$= \frac{84}{1000} * 10000 = 840\ kg/day$$

Corresponding COD consumed in sulphate reduction $= 840 * 0.67 = 562.8\ kg/d$

iii. COD available for methane production $= 6560 - 562.8 = 5997.2\ kg/d$

Methane produced at $27°C = 350\ L/kg\ COD\ removed$

Total methane produced/day $= 5997.2 * 0.350 * \frac{(273.15+27)}{273.15} = 2306.5\ m^3/d$

iv. Methane leaving as dissolved in effluent $= 0.028 * 10000 = 280\ m^3/day$

v. Usable methane $= 2306.5 - 280 = 2026.5\ m^3/d$

$$= \frac{2026.5\ m^3/d}{6560\ kg/d} = 0.309\ m^3/\ kg\ COD$$

▶ <u>Sulphates - Sulphides</u>:

Sulphates contained in influent will be reduced to sulphides. Assuming 85% removal of sulphates and their reduction to sulphides,

$$S^{2-} = 0.85 * \frac{105}{3} = 29.75\ mg/L$$

$$\frac{COD}{SO4} = \frac{820}{105} = 7.8 \text{ (favourable)}$$

Any H_2S gas present would be corrosive and would necessitate scrubbing of the gas with a NaOH solution prior to the supply of gas.

Residual sulphides remaining in the UASB, and therefore leaving in the effluent will have to be treated by aeration during post treatment.

▶ Gas Collector Design:

Provide gas collector width (i.e., centre to centre distance between two deflector beams) of 4 m. Hence number of gas collector $= \frac{24}{4} = 6\,no.$ (each 35 m length).

▶ Aperture Width:

Maximum upflow velocity through aperture is to be limited to 5 m/hr at peak flow.

Aperture area required $= \frac{peak\,flow\,m^3/hr}{permissible\ aperture\ velocity\ m/hr} = \frac{10000*2}{5*24} = 166.67\ m^2$

Aperture width $= \frac{aperture\ area}{length\ of\ aperture * no.of\ aperture} = \frac{166.67}{35*6} = 0.794\ m$

Taking it as approximately equal to 0.8 m.

Provide inclined plate of gas collector at 50° angle to the horizontal.

Horizontal width of plate,

P_h = (total gas collector width – aperture width – hood width + overlap of gas collector over hood width) / 2

$$= \frac{4.0 - 0.8 - 0.5 + 0.05}{2}$$

$$= 1.375\text{m}$$

P_v = vertical height of plate collector

$$P_v = \tan 50° \times P_h = 1.639\text{m}$$

$$P_v = \text{say } 1.64\text{m}$$

▶ Deflector Beam:

$Width\ of\ deflector\ beam = aperture\ width + 2(overlap) = 0.8 + 2 * 0.1 = 1\ m$

Height of deflector beam = 0.5 m

Check for total side water depth of 5m

Sludge bed height up to base of deflector beam $= 2.2\ m$

Deflector beam + space up to inclined plate = 0.75m

Collector plate height, P_v = 1.64 m

Vertical water depth in settler = 0.4 m

Free board at top $= 0.5\ m$

Total reactor height $= 5.5\ m$

▶ Settling Zone:

Cross sectional area of settler $= (0.8 * 1.64) + (0.4 * 3.2) + (1.64 * 1.2) = 4.56\, m^2$

$$Volume = area * length * no. = 4.56 * 35 * 6 = 957.6\, m^3$$

$$HRT = \frac{volume}{flow\ rate} = \frac{957.6}{10000\ (\frac{1}{24})} = 2.3\ hrs \text{ (acceptable)}$$

$$Surface\ loading = \frac{10000\ m^3/d}{settler\ width * length * no.} = \frac{10000}{3.2*35*6} = 14.88\ m^3/m^2/d \qquad \text{(acceptable)}$$

Final effluent should be suitable for land irrigation (BOD < 100 mg/L, SS < 100 mg/L, Sulphide < 2 mg/L)

▶ Sludge Withdrawal and Dewatering:

Sludge produced $= 315\ mg/L = 0.315\ kg/m^3$

At a sludge concentration of 7 % solids, the sludge volume $= \frac{0.315*10000}{0.07} = 45\ m^3/d$

Sludge withdrawal pipes may be provided at three different levels one at floor level, the second a height of 1 to 1.5 meter above the floor and third at a bottom of the settler (So that effluent can be used if necessary for flush-out of the sludge drains). The maximum load from any points on the floor of the reactor to sludge outlet should be less than 12 m. Each sludge drain is provided with sluice valve located in a chamber. The sludge drain are taken to sludge sump and hence to sludge drying beds.

▶ Sludge Drying Bed

Sludge drying time of 7 days followed by 2 to 3 days for sludge lifting before the next cycle commences, and assuming 45 m^3/d will be applied at 25 cm depth

Total sludge bed area required $= 45 * \frac{10}{0.25\ m} = 1800\ m^2$

Provide 12 beds of length 15 m and width 10 m. The beds will be formed with graded gravel and sand with open-jointed, glazed stoneware pipes at bottom to drain the filtrate back to the plant.

▶ Sludge chamber

At peak flow volume, $V = 45 * 2 = 90\ m^3/d$

$$A * h = 90$$

$$\frac{\pi}{4} * d^2 * h = 90$$

Assume h = 4 m, d = 5.35 m

Adopted sump (circular) size 5.5 m Ø, (4.0 + 0.6) deep + 0.3m FB

4.2.7 DESIGN OF FACULTATIVE TYPE AERATED LAGOON

Provide facultative aerated lagoon of 1 day detention time

BOD removal will occur from 70 mg/L to 40 mg/L (i.e. 42.8 % removal)

Oxygen demand $= 0.428\ (incoming\ BOD\ ultimate)$

$$= 0.428\ (1.47\ * 70\ g/m^3\ *\ 10000 * (\frac{1}{1000})$$

$$= 440\ kg/d\ = 18.3\ kg/h$$

Also, oxygen demand of 20 mg/L sulphides $= 20 * 2 = 40\ mg\ oxygen/L$

Kilogram of solids deposited from carryover @ 200 mg/L for ½ hour = 41.6 kg (oxygen demand @ 0.1 kg oxygen/kg solids = 4.16 kg oxygen/d $= 0.17\ kg/hr$)

Total oxygen required $= 18.3\ kg/hr + 16.7\ kg/hr + 0.17 = 35.2\ kg/hr$

Power Required = 25.1 kW

Power level $= \frac{25.1*1000}{10000} = 2.51\ W/m^2$

4.2.8 BRIEF DESCRIPTION OF UNITS OF UASB STP

Screen: Pre-treatment will be done in screens and grit chamber. For each stream of 10 MLD mechanical screens of 25 mm opening has been devised to screen the extraneous and foreign material. It is expected most of the unwanted material like polythene and other pouches will be trapped at the screen level.

Grit chamber: Grit in sewage consists of coarse particles of sand, ash and clinkers, eggshells, bone chips, and much inert material inorganic in nature. The specific gravity of grit is usually in the range of 2.4 to 2.65. Grit is non-putrescible and possesses a higher hydraulic subsidence value than organic solids. Hence separation of gritting material from organic solids is necessary.

UASB reactor: The size of reactor has been designed in such a way to accommodate the down ward distribution pipes carrying sewage to reactor and to collect the biogas produced by the biological action and allow sludge blanket formation in the reactor. The water height of the reactor has been designed keeping in view the height of digested sludge storage and height of active biomass called sludge blanket and solid liquid separation.

Sludge drying bed: The digested sludge will be sent to sludge sump by piping and finally pumped to sludge drying beds for drying and removal purpose. A sludge drying and disposal cycle of 10 days has been taken with a sludge application thickness of 25 cm on drying beds.

Filtrate sump: The filtrate of the sludge drying bed will be recycled to STP at the outlet of grit chamber before UASB reactors, by pumping at filtrate sump.

Effluent disposal: The final effluent of 10 MLD UASB STP will be disposed through gravity pipe to river nearby. It should also be considered to pump the treated waste water into the canal situated some distance from the STP site for irrigation purpose, if its feasibility is worked out in near future.

Resource recovery: Two bye products will be received after sewage treatment namely Biogas & dried sludge cakes. The biogas can be utilized for the generation of electricity by (dual fuel/gas) engine. It is expected that biogas shall produce more electricity and shall be utilized at the STP campus for operation of screens, de-gritting units, sludge pumps, aerator & lighting of campus and laboratory cum workshop. If biogas production is more than demand then the extra biogas can be sold to the market. The dried sludge can be utilized as manure for agriculture and horticulture purpose. Besides this it may also be used as land filling purpose.

Gas Utilization: The bio gas obtained by the UASB process is highly rich in methane content. In general, about 75% biogas is CH_4 and approximately 10 - 20% is Carbon dioxide and about 1% is. H_2S. H_2S gas is harmful for the process also. Hence H_2S scrubber has been proposed to get rid of H_2S. After analysing the actual production of gas, its best use shall be finalised.

CONCLUSION

It is hoped that the design of 10 MLD STP based on UASB will encourage environmental engineers to consider not only treating wastewaters in a controlled manner with the help of STP but also create better economic opportunities in the market as biogas can be used as a form of energy which is handy for underdeveloped and developing nations.

In Project Phase I, we have studied various sewage treatment processes and visited some sites. These processes will aid in the more effective disposal of wastewater which is beneficial to water bodies, aquatic life and living beings. We adopt our project titled as "**DESIGN OF 10 MLD SEWAGE TREATMENT PLANT BASED ON UPFLOW ANAEROBIC SLUDGE BLANKET (UASB) REACTOR WITH DESIGN OF UASB IN MS EXCEL**". In this Semester (Project Phase II), we are submitting the design of various components of UASB based STP with the automated UASB Design calculations in MS Excel through the attached QR code.

REFERENCES

- Manual on Sewerage and Sewage Treatment Systems Part A Engineering - 2013, CPHEEO.
- CPCB website (http://cpcb.nic.in/GeneralStandards.pdf).
- IS 3025 Methods of Sampling and Test (Physical and Chemical) for Water and Wastewater.
- Wastewater Treatment Concepts and Design Approach by G.L. Karia and R.A. Christian.
- Wastewater Treatment for Pollution Control and Reuse by Soli J. Arceivala.
- Environmental Engineering (Vol. II) Sewage disposal and air pollution Engineering a text book by S. K. Garg.
- NPTEL online courses Wastewater management.
- MoEF website.